User Guide for SAP Cloud Infrastructure

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SAP Cloud Infrastructure | SAP Customers and Partners

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User Guide for SAP Cloud Infrastructure

This guide provides information on how to optimally use SAP Cloud Infrastructure with the SAP Cloud Infrastructure dashboard.

Projects

Projects host your resources, allow role assignment to users, and control the user's authorizations for the project.

Creating Projects

Create a new project to host your cloud resources, enable user role assignments, and control resource access.

Prerequisites

- You're logged in to your domains home page.

  Example

  `https://dashboard.`*`<region>`*`.cloud.sap/`*`<domain>`*`/home`

Procedure

1. Choose **Create a New Project**.
2. Enter a name.
3. Enter a description.
4. Choose **Create project**.

Results

SAP Cloud Infrastructure automatically sets the available external networks.

Related Information

External Networks

Editing Project Information

Update the name and description of an existing project.

Prerequisites

- You're logged in to your project's home page.

  Example

  `https://dashboard.<region>.cloud.sap/<domain>/<project-name>/home`

Procedure

1. In the project information section, next to **Project**, choose **Edit**.

   A dialog box opens.
2. Enter a name.
3. Enter a description.
4. Choose **Update**.

Viewing the Project Wizard

View the project wizard to guide you through your project's initial setup steps.

Context

The project wizard automatically appears each time you enter the project until you complete all configurations within the wizard. These configurations include the following:

- Resource pooling
- Network configuration

Procedure

Replace the `/home` ending of the project URL with `/identity/project/wizard`.

Example

`https://dashboard.<region>.cloud.sap/<domain>/<project>/identity/project/wizard`

Identifying Resources to Clean Up Before Project Deletion

Identifying the resources associated with a project is a crucial step before deletion. All resources, such as servers, volumes, and networks, must be identified and cleaned up before you can delete a project.

Prerequisites

- You have the `admin` role.

  For more information, see Authorizations.

- You're logged in to your project's home page.

  Example

  `https://dashboard.<region>.cloud.sap/<domain>/<project-name>/home`

Procedure

In the project information section, next to **Delete Project**, choose **Check**.

A dialog box opens.

- If no resources are found, you can proceed to delete the project in the next step.
- If resources are found, they're listed. You must clean up the resources according to their delete order, starting with the lowest, before proceeding with the deletion.

Note

Backups aren't included in the resource reporting.

Next Steps

- Overview of Project Resources and Cleanup Procedures
- Deleting Projects

Overview of Project Resources and Cleanup Procedures

Overview of the resources that can be associated with a project and, if present, must be cleaned up before project deletion. You must clean up identified resources according to their delete order, starting with the lowest.

The table lists possible resources that can prevent project deletions. It also includes links to the relevant cleanup procedures for each resource:

Resource	Cleanup Procedures
Security Groups	Deleting Security Group Rules from Security Groups Deleting Security Groups
Ports	Detaching Interfaces from Server Instances Deleting Fixed IP Ports

Resource	Cleanup Procedures
Networks	Deleting Private Networks
Subnets	Deleting Subnets from Private Networks
Routers	Deleting Routers
Volume Snapshots	Deleting Volume Snapshots
Volumes	Detaching Volumes from Server Instances Deleting Volumes
Load Balancers	Deleting Pools Deleting Listeners Deleting Load Balancers
Object Storage	Deleting All Objects from an Object Storage Container Deleting Object Storage Containers
File System Storage	Deleting File Share Snapshots Deleting File Shares
Container Image Registry	Deleting Container Image Accounts
Key Manager	Deleting Secrets Deleting Secrets Containers

Deleting Projects

Permanently delete a project you no longer need.

Prerequisites

- You have the `admin` role.

  For more information, see Authorizations.

- You're logged in to your project's home page.

- You've identified and cleaned up all project resources.

  For more information, see Identifying Resources to Clean Up Before Project Deletion.

- The project must not contain any sub-projects. If a sub-project exists, you must delete the sub-project before proceeding with the deletion of the main project.

Context

⚠ Caution

When deleting the project, there's no option to roll back.

Procedure

1. In the project information section, next to **Delete Project**, choose **Check**.

   A dialog box opens.

2. Choose **Delete**.

Technical Prerequisites

[Supported Browsers](#)
Get an overview of supported browsers and browser settings.

Supported Browsers

Get an overview of supported browsers and browser settings.

SAP Cloud Infrastructure supports the following browsers:

Browser	Version	Additional Information
Google Chrome	latest version	We make every effort to fully test and support the latest versions as they are released. For additional system requirements, see your web browser documentation.
Safari	latest version	We make every effort to fully test and support the latest versions as they are released. For additional system requirements, see your web browser documentation.
Firefox	latest version	We make every effort to fully test and support the latest versions as they are released. For additional system requirements, see your web browser documentation.
Microsoft Edge	latest version	We make every effort to fully test and support the latest versions as they are released. For additional system requirements, see your web browser documentation.

Project Services

[Compute](#)

Manage compute resources to ensure the processing power needed for running applications and handling various workloads.

Networking and Loadbalancing

Manage network connectivity and traffic distribution to optimize application performance and reliability.

Storage

Manage storage capacity to save and manage your data.

Authorizations

Manage user and group access, and handle sensitive data like encryption keys, certificates, and passphrases with the key manager.

API Access

Manage API authentication and configuration using endpoint parameters.

Capacity and Metrics

Track resource usage and quotas. Explore metrics and audit logs to monitor and analyze your cloud environment.

Compute

Manage compute resources to ensure the processing power needed for running applications and handling various workloads.

Servers

Create, modify, and manage server instances. Perform actions such as resizing, starting, stopping, and rebooting instances. Attach or detach resources like floating IPs and interfaces, and organize servers with tags.

Volumes and Volume Snapshots - Block Storage

Create, modify, and manage volumes and volume snapshots. Perform actions such as attaching volumes to server instances, taking snapshots, and deleting volumes or snapshots when no longer needed. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

Server Images and Server Image Snapshots

Manage server images and snapshots by sharing images with other projects, and launching new instances. Perform actions like setting images to private or shared visibility and deleting images when no longer needed.

SSH Key Pairs

Enable secure access to your server instances. A key pair consists of a private key, which you keep safe on your local machine, and a public key, which you upload to your cloud account.

Flavors

A flavor is a predefined hardware configuration for a server, defining the size of a virtual server to be launched, including its compute, memory, and storage capacities.

Servers

Create, modify, and manage server instances. Perform actions such as resizing, starting, stopping, and rebooting instances. Attach or detach resources like floating IPs and interfaces, and organize servers with tags.

Creating Server Instances

Create new server instances.

Editing Server Instance Names

Update the name of an existing server instance.

Adding Tags to Server Instances

Add tags to server instances to organize them by specific categories, such as environment or department. Tags make it easier to manage and filter your servers.

Attaching Floating IPs to Server Instances

Attach floating IPs to server instances to enable external network connections. This process binds the instances' fixed IPs to the floating IPs, allowing traffic between the server instances and external networks.

Detaching Floating IPs from Server Instances

Detach a floating IP to reassign it to another server instance or remove public access when no longer needed.

Attaching Interfaces to Server Instances

Attach an interface to create a port to your server instance.

Detaching Interfaces from Server Instances

Detach an interface to remove a port from your server instance.

Resizing Server Instances

Change the size of a server instance by changing the flavor.

Creating Server Image Snapshots

Create a snapshot of your running server instance. Later, you can use this snapshot as a source image to create new instances from it.

Starting Server Instances

Start a stopped server instance to bring it back online.

Stopping Server Instances

Stop a running server instance to save resources or perform maintenance.

Rebooting Server Instances

Reboot a running server instance to perform a graceful shut down and restart of the server instance.

Pausing Server Instances

Pause a running server instance to temporarily suspend operations.

Suspending Server Instances

Suspend a server instance, similar to hibernation, to free up resources while preserving its state.

Deleting Server Instances

Permanently delete a server instance you no longer need.

Server Group Policies

Get an overview of affinity and anti-affinity policies that determine how virtual machines are scheduled on physical hosts.

Creating Server Instances

Create new server instances.

Prerequisites

- You have the `admin` role.

  For more information, see Authorizations.

- You have generated an SSH key pair and uploaded your public key.

  For more information, see Uploading an SSH Public Key.

- You know which flavor you want to use. The flavor defines the root disk size, RAM, and number of CPU's for the server instance.

  For more information, see Flavors.

- You know the server image you want to use.

- You have a security group.

  For more information, see Security Groups.

- You have a private network, which is required to allocate an initial fixed IP.

For more information, see Private Networks.

- **Optional:** You have created subnets in your private networks to segment traffic.

  For more information, see Creating Subnets in Private Networks.

- **Optional:** You have reserved a fixed IP.

  For more information, see Reserving Fixed IPs.

- **Optional:** You have prepared a cloud-init script for the **User Data** field to customize instance setup during creation. This can be used for various purposes, such as injecting multiple SSH public keys.

  For more information, see Creating a user-data File to Inject Multiple SSH Public Keys.

Procedure

1. Choose **Compute** > **Servers**.
2. Choose **Create New**.

   A dialog box opens.
3. Enter a name in lower case letters.
4. Select an availability zone that has enough available space.

   If the availability zone doesn't have enough space, the server instance can't be created. You'll get the error:
   `Error: No valid host was found.`

   > **Note**
   >
   > If you plan to attach a volume, ensure both the volume and server instance are in the same availability zone.

5. Select the security group to manage ingress and egress traffic.
6. Select your SSH key pair.
7. Select a flavor.
8. **Optional:** To use a volume as the root disk and define its size in gigabytes, select the **Custom root disk** checkbox. If you leave the checkbox deselected, the server uses the ephemeral root disk size defined by the selected flavor.
9. Select a server image.
10. Select the private network.
11. **Optional:** To customize network settings, choose **Toggle advanced network options**.

    These settings cater to advanced use cases that require more control over the network configuration of new server instances. You can only access these options by first creating a port in the selected subnet or with the selected fixed IP. Then, add the port to the server instance during its creation.

    > **Caution**
    >
    > Ports created specifically for this purpose won't be deleted when you delete the server instance. Floating IPs attached to these ports won't be released automatically either. We recommend using these options only if necessary for your use case. Ensure you clean up the ports manually afterward.

    a. Select the subnet from which the fixed IP is chosen.

       If you leave the **Subnet** field empty, the system selects any subnet.

    b. Enter a previously reserved fixed IP address or a new one.

       If you leave the **Server fixed IP** field empty, DHCP automatically selects an IP address for you.

12. **Optional:** Paste cloud-init YAML content into the **User Data** field to configure server instance behavior during first boot.

    Example

    Inject multiple SSH public keys:

    Sample Code

    ```
    #cloud-config
    ssh_authorized_keys:
    - "ssh-rsa AAAAB3...key1"
    - "ssh-rsa AAAAB3...key2"
    ```

    Ensure the content is properly formatted and the block is placed correctly if you add additional cloud-init directives such as `run-cmd`.

13. Choose **Create**.

Results

- The server instance status changes as follows:
  - **Scheduling**: The system prepares resources for the server instance.
  - **Spawning**: The server instance is being created.
  - **Active**: The server instance has been created and is now running.
- The power state changes to **Running** once the server instance is active.
- The server instance is equipped with one port upon instantiation.
- The system sets a random password for the `admin` user and encrypts it using the provided public key.
- The activity is recorded as an event in the audit log.

Next Steps

- Retrieve the password using the nova-password tool.

  For more information, see https://github.com/sapcc/nova-password .
- **Optional:** Attaching Floating IPs to Server Instances

Related Information

Fixed IPs / Ports
Audit

Editing Server Instance Names

Update the name of an existing server instance.

Prerequisites

- You have the `member` role.

  For more information, see Authorizations.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Edit Name**.

   A dialog box opens.
3. Enter a name.

   Ensure that it doesn't end with a dot or a number.
4. Save your entries.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Adding Tags to Server Instances

Add tags to server instances to organize them by specific categories, such as environment or department. Tags make it easier to manage and filter your servers.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Edit Tags**.

   A dialog box opens.
3. Choose **+ New Tag**.
4. Enter a tag.
5. Save your entries.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Attaching Floating IPs to Server Instances

Attach floating IPs to server instances to enable external network connections. This process binds the instances' fixed IPs to the floating IPs, allowing traffic between the server instances and external networks.

Prerequisites

- You have a server instance without a floating IP attached. Each server instance can have only one floating IP assigned at a time.

  For more information, see Creating Server Instances.

- Your network is connected to a router to route external traffic.

  For more information, see Creating Routers.

- You have a floating IP.

  For more information, see Creating Floating IPs.

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Attach Floating IP**.

   A dialog box opens.

3. Select the interface IP.
4. Select a floating IP that is part of the external subnet you want to use.
5. Choose **Attach**.

Results

- The server instance has a floating IP attached.

- The activity is recorded as an event in the audit log.

Related Information

Audit

Detaching Floating IPs from Server Instances

Detach a floating IP to reassign it to another server instance or remove public access when no longer needed.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You have a floating IP attached to a server instance.

For more information, see Attaching Floating IPs to Server Instances.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Detach Floating IP**.

   A dialog box opens.
3. Select the IP address.
4. Choose **Detach**.

Results

- The floating IP is no longer attached to the server instance.
- The floating IP remains allocated to your project and is ready for you to reattach or release.
- The activity is recorded as an event in the audit log.

Next Steps

- Attaching Floating IPs to Server Instances
- Releasing Floating IPs

Related Information

Audit

Attaching Interfaces to Server Instances

Attach an interface to create a port to your server instance.

Prerequisites

- If you're using VMware, you can attach a maximum of 10 ports to a single server instance due to a VMware limitation.

Context

Ports (also known as Interfaces or NICs) are the interface between an instance and the network the instance is connected to. All kinds of different networking quantifiers are related to ports directly such as IPs, MACs, security groups, and so on.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Attach Interface**.

   A dialog box opens.
3. Select the network.
4. Select the subnet.

5. **Optional:** Enter the IP address that you want the interface to have, or leave empty for DHCP allocation.
6. **Optional:** Choose the security groups that you want this interface to have.

   Other interfaces on the same server can have different security groups.

   When you add or remove a security group to or from a server, the change applies to all of its interfaces.
7. Choose **Attach**.

Next Steps

- For more granular control over your instances' networking, manage the ports directly.

  For more information, see [Fixed IPs / Ports].
- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Detaching Interfaces from Server Instances

Detach an interface to remove a port from your server instance.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Detach Interface**.

   A dialog box opens.
3. Select the IP address of the interface you want to detach.

   Note

   If you choose an alias of another IP on an interface, the interface itself alongside all its IPs will be detached from the server.
4. Choose **Detach**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Fixed IPs / Ports]
[Audit]

Resizing Server Instances

Change the size of a server instance by changing the flavor.

Prerequisites

- The new flavor does not contain a smaller root disk size.

  For more information, see Flavors.

- Ensure the availability zone has enough available space. During a resize, both the old and new sizes will consume the full amount of memory and other resources.

  For more information, see Resource Management.

Context

Resizing a server is a technically complex operation and can fail for various reasons. In many cases, a better approach is to create a new server in the desired size and delete the old one.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Resize**.

   A dialog box opens.
3. Select the new flavor.
4. Choose **Resize**.

   The server instance status changes to **Resize migrating**.

   The server instance is resized. If the server instance was running, it is shut down and restarted during the process. If it was shut off, it remains in **Shutoff** status.

   After the operation completes, the server instance status changes to **Verify resize**.
5. From the gear dropdown menu for the server instance, choose **Confirm Resize**.

   Note

   If you do not want to confirm the resize, you can choose **Revert Resize** instead. This cancels the operation and restores the server instance to its original flavor and previous status.

Results

- After confirming the resize, the server instance returns to its previous status with the new flavor applied.
- If the resize is reverted, the instance returns to its original flavor and previous status.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Creating Server Image Snapshots

Create a snapshot of your running server instance. Later, you can use this snapshot as a source image to create new instances from it.

Prerequisites

- You have the following roles:
  - `image_admin`
  - `objectstore_admin`

  For more information, see Authorizations.
- The server instance is in power state **Running**.
- You have sufficient object storage quota. Otherwise the process fails silently.

  For more information, see Resource Management.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Create Snapshot**.

   A dialog box opens.
3. Enter a name for the snapshot.
4. Choose **Create Snapshot**.

   The snapshot status is shown in the **Status** column. The operation is completed once the status changes from **Image uploading** to **Active**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Server Images and Server Image Snapshots
Disabling Volume Shadow Copy Service Application Quiescing
Audit

Starting Server Instances

Start a stopped server instance to bring it back online.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Start**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Stopping Server Instances

Stop a running server instance to save resources or perform maintenance.

Prerequisites

- The server instance is in power state **Running**.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Stop**.
3. Confirm the dialog box.

Results

- The server instance is in power state **Shut down** and status **Shutoff**.
- Allocated resources (such as vCPU, RAM, and storage) remain reserved.
- The server instance continues to count toward the project's resource usage.
- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Rebooting Server Instances

Reboot a running server instance to perform a graceful shut down and restart of the server instance.

Prerequisites

- The server instance is in power state **Running**.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Reboot**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Pausing Server Instances

Pause a running server instance to temporarily suspend operations.

Prerequisites

- The server instance is in power state **Running**.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Pause**.
3. Confirm the dialog box.

Results

- Allocated resources (such as vCPU, RAM, and storage) remain reserved.
- The server instance continues to count toward the project's resource usage.
- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Suspending Server Instances

Suspend a server instance, similar to hibernation, to free up resources while preserving its state.

Prerequisites

- The server instance is in power state **Running**.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Suspend**.
3. Confirm the dialog box.

Results

- Allocated resources (such as vCPU, RAM, and storage) remain reserved.
- The server instance continues to count toward the project's resource usage.
- The activity is recorded as an event in the audit log.

Next Steps

Starting Server Instances

Related Information

Audit

Deleting Server Instances

Permanently delete a server instance you no longer need.

Procedure

1. Choose **Compute** > **Servers**.
2. From the gear dropdown menu for the server instance, choose **Terminate**.
3. Confirm the dialog box.

Results

- The server instance is now permanently deleted.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Server Group Policies

Get an overview of affinity and anti-affinity policies that determine how virtual machines are scheduled on physical hosts.

Note
To apply placement policies, use the CLI.

Policy	Description
Affinity Policy	• The system schedules all server instances on the same physical host.

Policy	Description
	• If scheduling fails, instances don't get created, and the system enters an error state. • No network equipment between server instances, reducing possible points of failure, latency, and bandwidth bottlenecks. • If the host fails, all instances in the group fail together.
Soft-Affinity Policy	• The system tries to schedule instances on the same host, but it doesn't guarantee placement. • If scheduling can't be completed, instance creation still proceeds. • Instances are most likely to be placed in the same rack, but not necessarily on the same host.
Anti-Affinity Policy	• The system schedules instances on different building blocks (racks). • If scheduling cannot meet this requirement, instance creation will fail, and the system will return an error state. • If a host fails, only one instance in the group is affected. • Additional details: A "host" in this context refers to a cluster, so instances may run on different hypervisors. If root disks are not created via volumes, they are stored on two separate storage backends for redundancy.
Soft-Anti-Affinity Policy	• The system tries to schedule instances across different building blocks but does not enforce this placement. • Within a building block: Cluster rules ensure that instances in the same server group do not run on the same hypervisor. If root disks are not created via volumes, they are stored on two separate backends

Related Information

Command-Line Guide for SAP Cloud Infrastructure

Volumes and Volume Snapshots - Block Storage

Create, modify, and manage volumes and volume snapshots. Perform actions such as attaching volumes to server instances, taking snapshots, and deleting volumes or snapshots when no longer needed. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

Block Storage Configuration Maximum

Get an overview of the maximum sizes for block storage.

Resource	Maximum Size
Block Storage Volume	10 TB
Data Disks	59 disks

Volume Affinity and Anti-Affinity

Volume affinity and anti-affinity are placement policies that control where a new volume is physically placed relative to an existing volume during creation. These policies optimize volume distribution for performance or high availability and are applied using scheduler hints.

Note

To apply placement policies, use the CLI.

Placement Policies

Placement Policy	Description
Affinity	Places a new volume on the same host and storage backend as a specified existing volume using the `same_host` scheduler hint.
Anti-affinity	Places a new volume on a different host and storage backend from a specified volume using the `different_host` scheduler hint.

Related Information

[Command-Line Guide for SAP Cloud Infrastructure]

Creating Volumes

Create new volumes to allocate space for your server instances.

Prerequisites

- You have the `volume_admin` role.

  For more information, see [Authorizations].

- You have a server instance.

  For more information, see [Creating Server Instances].

- The volume must be created in the same availability zone as the server instance you want to attach it to.

- To apply placement policies such as affinity or anti-affinity, create volumes using the CLI.

  For more information, see [Volume Affinity and Anti-Affinity].

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. Choose **Create New**.

   A dialog box opens.

3. Enter a name.
4. Enter a description.
5. Enter the size in GB.
6. To use the volume as root disk (bootable volume), select the **bootable** checkbox.
7. Select the volume type. You have the following options:

- premium_ssd

    The default volume type, backed by SSD disks, providing the highest performance.

- standard_hdd

    A more economical choice for less demanding storage needs than premium_ssd. It has slower volume attach times but is cost-effective, with a minimum volume size of 500 GB.

8. Select the availability zone where the existing server instance is located.
9. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

[Attaching Volumes to Server Instances]

Related Information

[Audit]

Editing Volumes

Update the name and description of an existing volume.

Prerequisites

- You have the `volume_admin` role.

    For more information, see [Authorizations].

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Edit**.

    A dialog box opens.
3. Enter a name.
4. Enter a description.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Cloning Volumes

Create exact copies of existing volumes for backup or replication purposes.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Compute** > **Volumes & Snapshots** .
2. From the gear dropdown menu for the volume, choose **Clone Volume**.

   A dialog box opens.
3. **Optional:** Adjust the pre-filled information.
4. Choose **Clone**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Identifying Orphaned Volumes

Show only storage volumes that are in the status **available** to identify unused capacity. Volumes in this status are not attached to any server and are potential candidates for removal.

Prerequisites

- You have the `volume_viewer` role.

  For more information, see Authorizations.

Context

Storage volumes in status **available** are not attached to any server. These volumes are not in active use. The **Updated at** timestamp shows the last change to the volume, such as attaching or detaching it. The longer a volume remains in the available state without updates, the more likely it is no longer needed.

Procedure

1. Choose **Compute** > **Volumes & Snapshots** .
2. Select the **Show only available volumes** checkbox.
3. Review the **Updated at** column to see when each volume was last changed.

4. Calculate how many days have passed since the last update to determine how long the volume has been orphaned.

Results

You see all volumes that are not attached to a server and how long they have been idle. Use this data to decide whether to remove unused volumes and free up resources.

Attaching Volumes to Server Instances

Link volumes to available server instances to expand storage capacity or access data.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

- You have a server instance.

  For more information, see Creating Server Instances.

- The volume is in status **available**.
- The volume is in the same availability zone as the server instance.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Attach**.

   A dialog box opens.

3. Specify the server instance to which you want to attach the volume. You have the following options:
   - Enter a server ID.
   - Select a server instance.
4. Choose **Attach**.

Results

- The activity is recorded as an event in the audit log.
- The volume is in status **in-use**.

Next Steps

- Initialize and mount the volume for use.

Related Information

Audit

Detaching Volumes from Server Instances

Detach volumes from server instances to stop access or free up resources.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

- The volume is in status **in-use**.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Detach**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

- The volume is in status **available**.

Related Information

Audit

Deleting Volumes

Delete volumes that you no longer need.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

- The volume is in status **available**.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Delete**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Uploading Volumes to Server Images

Convert a volume into a server image for easier storage, sharing, and access in future instances.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

- The volume is in status **available**.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Upload to Image**.

   A dialog box open.

3. Enter the server image name.
4. **Optional:** Select the disc format:
   - **vmdk**
   - **raw**
5. **Optional:** Select the container format
6. **Optional:** Select the visibility:
   - **private**
   - **shared**
   - **public**
7. **Optional:** To restrict operations on specific server image properties to particular user roles, select the **Protect this image** checkbox.
8. Choose **Clone**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Server Images and Server Image Snapshots

Related Information

Audit

Disabling Volume Shadow Copy Service Application Quiescing

Disable Volume Shadow Copy Service (VSS) application quiescing in VMware Tools by editing the VMware Tools configuration file on the virtual machine. Disabling VSS application quiescing can help when there are issues with backup solutions or snapshot consistency.

Prerequisites

- Operating System: Applicable only to Windows virtual machines.
- VMware Tools is installed on the Windows virtual machine.

  For more information, see VMware Tools Documentation .

Procedure

1. Open the VMware Tools configuration file in a text editor.

   You can find the file in the following directory: `C:\ProgramData\VMware\VMware Tools\tools.conf`

   If the `tools.conf` file doesn't exist, create the file.
2. Add the following lines to disable VSS application quiescing:

   Code Syntax

   ```
   [vmbackup]
   vss.disableAppQuiescing = true
   ```
3. Save and close the file.
4. To restart the VMware Tools service, press Win + R .
5. Enter `services.msc`.
6. Press Enter.
7. Right-click on **VMware Tools** and choose **Restart**.

Results

VSS application quiescing is disabled. This prevents VMware Tools from using VSS to quiesce applications during snapshots.

Related Information

Creating Volume Snapshots
Creating Server Image Snapshots

Creating Volume Snapshots

Take snapshots of volumes to capture their state for backup or restoration.

Prerequisites

- You have the `volume_admin` role.

For more information, see Authorizations.

- The volume is in status **available**.
- The volume is not attached to a server instance.

  For more information, see Detaching Volumes from Server Instances.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume, choose **Create Snapshot**.

   A dialog box opens.
3. **Optional:** Adjust the pre-filled information.
4. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating Volumes from Volume Snapshots

Related Information

Disabling Volume Shadow Copy Service Application Quiescing
Audit

Creating Volumes from Volume Snapshots

Create new volumes based on existing volume snapshots to restore or replicate data.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

- You have a volume snapshot.

  For more information, see Creating Volume Snapshots.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. Choose the **Volume Snapshots** tab page.
3. From the gear dropdown menu for the volume snapshot, choose **Create Volume**.

   A dialog box opens.

4. **Optional:** Adjust the pre-filled information.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing Volume Snapshots

Update the name and description of an existing volume snapshot.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. From the gear dropdown menu for the volume snapshot, choose **Edit**.

   A dialog box opens.
3. Enter a name.
4. Enter a description.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Volume Snapshots

Permanently delete a volume snapshot you no longer need.

Prerequisites

- You have the `volume_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Compute** > **Volumes & Snapshots**.
2. Choose the **Volume Snapshots** tab page.
3. From the gear dropdown menu for the volume snapshot, choose **Delete**.
4. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Server Images and Server Image Snapshots

Manage server images and snapshots by sharing images with other projects, and launching new instances. Perform actions like setting images to private or shared visibility and deleting images when no longer needed.

Launching Instances
Create server instances from a snapshot or server image.
Sharing Server Images with other Projects
Share server images with other projects in the same region using access control.
Accepting an Access Control Request
Allow access to a shared server image by accepting an access control request.
Setting Server Images to Private Visibility
Restrict access to a server image by setting it to private visibility.
Setting Server Images to Shared Visibility
Make a server image accessible to specific users or projects by setting it to shared visibility.
Deleting Private Server Images
Permanently delete a server image you no longer need.

Launching Instances

Create server instances from a snapshot or server image.

Prerequisites

- You have the `admin` role.

  For more information, see Authorizations.

- You have a server image snapshot or a server image.

  For more information, see:

  - Creating Server Image Snapshots

- Uploading Volumes to Server Images

- You have generated an SSH key pair and uploaded your public key.

  For more information, see Uploading an SSH Public Key.

- You know which flavor you want to use. The flavor defines the root disk size, RAM, and number of CPU's for the server instance.

  For more information, see Flavors.

- You know the server image you want to use.

- You have a security group.

  For more information, see Security Groups.

- You have a private network, which is required to allocate an initial fixed IP.

  For more information, see Private Networks.

- **Optional:** You have created subnets in your private networks to segment traffic.

  For more information, see Creating Subnets in Private Networks.

- **Optional:** You have reserved a fixed IP.

  For more information, see Reserving Fixed IPs.

- **Optional:** You have prepared a cloud-init script for the **User Data** field to customize instance setup during creation. This can be used for various purposes, such as injecting multiple SSH public keys.

  For more information, see Creating a user-data File to Inject Multiple SSH Public Keys.

Procedure

1. Choose **Compute** > **Server Images & Snapshots**.
2. From the gear dropdown menu for the snapshot or server image, choose **Launch Instance**.

   A dialog box opens.
3. Enter a name in lower case letters.
4. Select an availability zone that has enough available space.
5. Select the security group to manage ingress and egress traffic.
6. Select your SSH key pair.
7. Select a flavor.
8. **Optional:** To use a volume as the root disk and define its size in gigabytes, select the **Custom root disk** checkbox. If you leave the checkbox deselected, the server uses the ephemeral root disk size defined by the selected flavor.
9. Select a server image.
10. Select the private network.
11. **Optional:** To customize network settings, choose **Toggle advanced network options**.

    These settings cater to advanced use cases that require more control over the network configuration of new server instances. You can only access these options by first creating a port in the selected subnet or with the selected fixed IP. Then, add the port to the server instance during its creation.

    ⚠ **Caution**

Ports created specifically for this purpose won't be deleted when you delete the server instance. Floating IPs attached to these ports won't be released automatically either. We recommend using these options only if necessary for your use case. Ensure you clean up the ports manually afterward.

a. Select the subnet from which the fixed IP is chosen.

   If you leave the **Subnet** field empty, the system selects any subnet.

b. Enter a previously reserved IP address or a new one.

   If you leave the **Server fixed IP** field empty, DHCP automatically selects an IP address for you.

12. **Optional:** Paste cloud-init YAML content into the **User Data** field to configure server instance behavior during first boot.

Example

Inject multiple SSH public keys:

Sample Code

```
#cloud-config
ssh_authorized_keys:
- "ssh-rsa AAAAB3...key1"
- "ssh-rsa AAAAB3...key2"
```

Ensure the content is properly formatted and the block is placed correctly if you add additional cloud-init directives such as `run-cmd`.

13. Choose **Create**.

Results

- The system sets a random password for the `admin` user and encrypts it using the provided public key.
- The activity is recorded as an event in the audit log.

Next Steps

- Retrieve the password using the nova-password tool.

  For more information, see https://github.com/sapcc/nova-password .
- **Optional:** Attaching Floating IPs to Server Instances

Related Information

Creating Server Instances

Fixed IPs / Ports

Audit

Sharing Server Images with other Projects

Share server images with other projects in the same region using access control.

Prerequisites

- The server image is set to shared visibility.

  For more information, see Setting Server Images to Shared Visibility.

- The target project is in the same region.

Procedure

1. Choose **Compute** > **Server Images & Snapshots**.
2. From the gear dropdown menu for the server image, choose **Access Control**.

   A dialog box opens.

3. Choose **+**.

   A dialog box opens.

4. Enter a target project name or project ID.
5. Choose **Add**.

   The added target project is in status **pending**.

6. Notify the project owner or a user with access to the entered target project that they need to accept the access control request.

   Once accepted, the users in the target project will have access to the shared server image.

Related Information

Accepting an Access Control Request

Accepting an Access Control Request

Allow access to a shared server image by accepting an access control request.

Prerequisites

- A private image has been shared.

  For more information, see Sharing Server Images with other Projects.

Procedure

1. Choose **Compute** > **Server Images & Snapshots**.
2. Choose the **Suggested** tab page.
3. From the gear dropdown menu for the image, choose **Accept**.

Results

- The image disappears from the **Suggested** tab and shows up on the **Available** tab when you select the **shared** radio button.
- In the source projects **Access Control** screen, the status of the target project is now set to accepted.

Setting Server Images to Private Visibility

Restrict access to a server image by setting it to private visibility.

Prerequisites

- The server image is set to shared visibility.

Procedure

1. Choose **Compute** > **Server Images & Snapshots**.
2. Use the filter to show only server images that are set to shared visibility.
3. From the gear dropdown menu for the server image, choose **Set to private**.

Setting Server Images to Shared Visibility

Make a server image accessible to specific users or projects by setting it to shared visibility.

Prerequisites

- The server image is set to private visibility.

Procedure

1. Choose **Compute** > **Server Images & Snapshots**.
2. Use the filter to show only server images that are set to private visibility.
3. From the gear dropdown menu for the server image, choose **Set to shared**.

Next Steps

[Sharing Server Images with other Projects]

Deleting Private Server Images

Permanently delete a server image you no longer need.

Prerequisites

- You have the following roles:
  - `image_admin`
  - `objectstore_admin`

  The server images service uses the object storage service as its backend. You need both roles to delete an image and its stored data.

  For more information, see [Authorizations].

Procedure

1. Choose Compute > Server Images & Snapshots.
2. Select the **private** radio button.
3. From the gear dropdown menu for the server image, choose **Delete**.
4. Confirm the dialog box.

Results

The image is deleted. If the image has been shared with other projects, it is removed from all projects that had access to this image.

SSH Key Pairs

Enable secure access to your server instances. A key pair consists of a private key, which you keep safe on your local machine, and a public key, which you upload to your cloud account.

[Uploading an SSH Public Key](#)

Enable secure SSH access to your server instances by uploading your public SSH key.

[Creating a user-data File to Inject Multiple SSH Public Keys](#)

Create a `user-data.yaml` file to inject multiple SSH public keys into a server instance during creation. The keys are added using `cloud-init`.

[Deleting an Uploaded SSH Public Key](#)

Remove an existing SSH public key from your registered key pairs to prevent it from being used for new server instances.

Uploading an SSH Public Key

Enable secure SSH access to your server instances by uploading your public SSH key.

Prerequisites

- You have generated an SSH key pair on your local machine.
- You have access to your public SSH key in OpenSSH format.

Example

- macOS and Linux: `~/.ssh/id_rsa.pub` or `~/.ssh/id_ed25519.pub`
- Windows: `C:\Users\<your name>\.ssh\id_rsa.pub` or `C:\Users\<your name>\.ssh.id_ed25519.pub`

Procedure

1. From the user menu in the top header bar, choose **Key Pairs**.

   The key pairs UI page opens, showing any existing keys.
2. Choose **Create new**.

A dialog box opens.

3. Enter your name.
4. Paste your public key in OpenSSH format.
5. Choose **Save**.

Results

- Your public key is now registered. When creating a new server instance, you can select it from the **Key Pair** dropdown to enable secure SSH access using the matching private key.
- The activity is recorded as an event in the audit log.

Next Steps

- [Creating Server Instances]
- [Launching Instances]

Related Information

[Audit]

Creating a user-data File to Inject Multiple SSH Public Keys

Create a `user-data.yaml` file to inject multiple SSH public keys into a server instance during creation. The keys are added using `cloud-init`.

Prerequisites

- You have access to the SSH public keys you want to inject.

Procedure

1. On your local machine, create a file named `user-data.yaml`.
2. Add the following content to the file. Replace the example keys with your actual SSH public keys:

   ### Code Syntax

   ```
   #cloud-config
   ssh_authorized_keys:
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC0...key1"
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC1...key2"
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC2...key3"
   ```

   Each key must be on a single line. Use correct indentation.
3. Save the file.

Next Steps

- [Creating Server Instances]

- Launching Instances

Deleting an Uploaded SSH Public Key

Remove an existing SSH public key from your registered key pairs to prevent it from being used for new server instances.

Context

Deleting an uploaded SSH public key from the dashboard removes it from the list of available key pairs for creating or launching new server instances. This action does not delete the corresponding private key stored on your local machine. The private key remains on your device unless you remove it manually.

Additionally, any existing server instances that were created with this key pair remain accessible using the corresponding private key unless you manually remove the key from the server's `~/.ssh/authorized_keys` file.

Procedure

1. From the user menu in the top header bar, choose **Key Pairs**.

   The key pairs UI page opens, showing any existing keys.
2. From the gear dropdown menu for the key pair, choose **Delete**.
3. Confirm the dialog box.

Results

- The public key is removed from your registered key pairs. It no longer appears as an option when creating or launching new server instances.

- The activity is recorded as an event in the audit log.

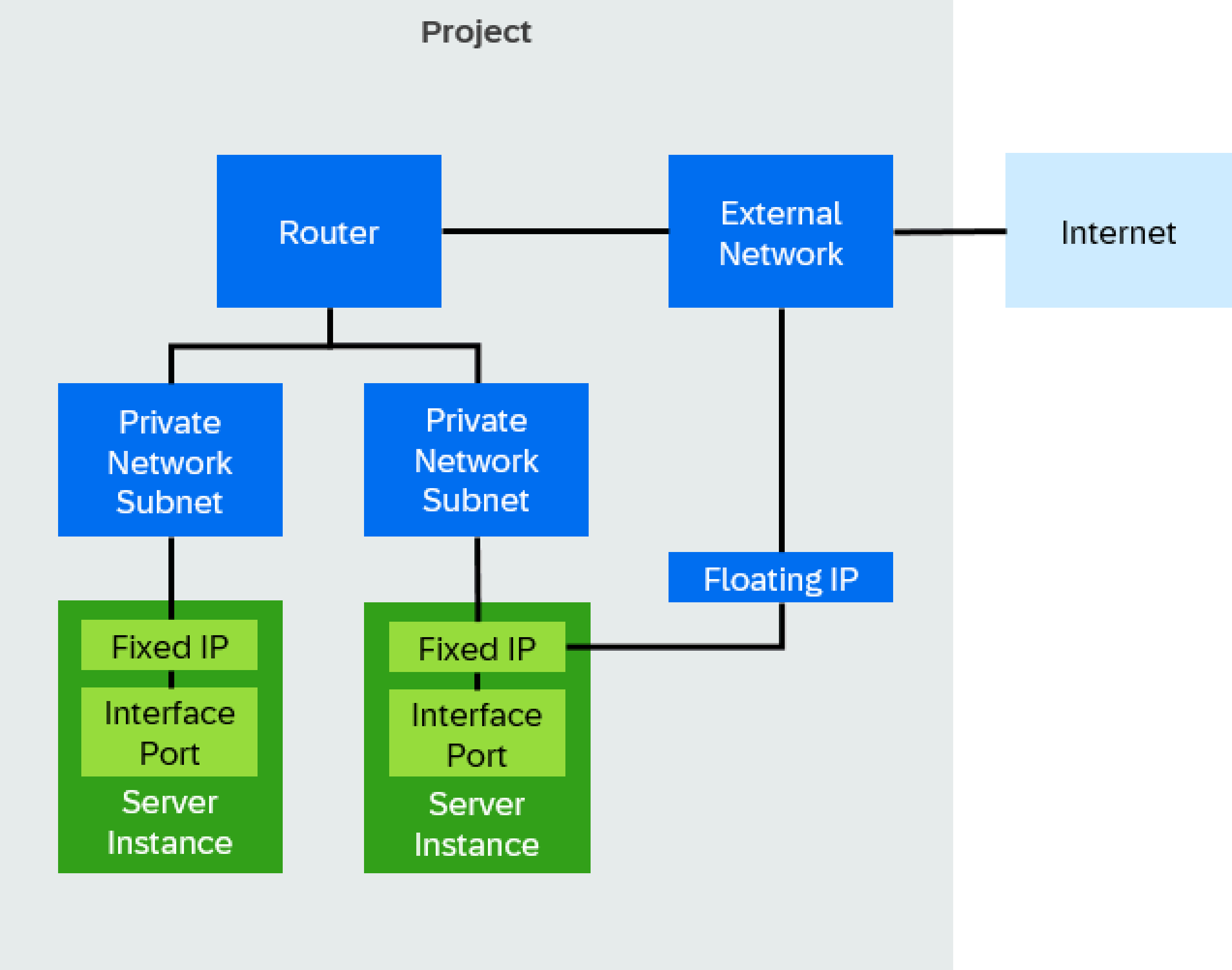
Related Information

Audit

Networking and Loadbalancing

Manage network connectivity and traffic distribution to optimize application performance and reliability.

This image is interactive. Hover over each area for a description.



Please note that image maps are not interactive in PDF outputs.

[Supported Network Connections](#)

Get an overview of supported network connections.

[Network Traffic Rules](#)

Get an overview of network traffic that is always enabled, regardless of security groups applied.

[External Networks](#)

Projects need at least one external network for an Internet connection. Without these networks, users can't access instances in the private network, leading to isolation. Domain owners create and manage external networks and their associated subnets.

[Private Networks](#)

Private networks isolate resources within a project and assign internal IP addresses for communication. Create, edit, and delete private networks, manage subnets, and share private networks with other projects.

[Routers](#)

A router connects subnets and external networks, enabling communication within your infrastructure. Manage router settings, configure routers, and view your router topology.

[Border Gateway Protocol VPN](#)

Border gateway protocol (BGP) VPNs are virtual private networks that use BGP to exchange routing information between connected networks. Manage BGP VPNs by creating them, adding or removing routers, sharing them with other projects, and deleting them when they are no longer needed.

[Fixed IPs / Ports](#)

Fixed IPs are IPv4 addresses that DHCP assigns to server instances (VMs). A fixed IP acts as a port on the private network and connects to the NIC of a server instance.

[Floating IPs](#)

Floating IPs (FIPs) are static IPs that belong to a project rather than a specific server instance. You can assign them to any server instance for SSH, RDP, or HTTP access. FIPs are created from the IPv4 pool of a subnet.

[Security Groups](#)

Manage security groups that act as virtual firewalls to control incoming (ingress) and outgoing (egress) network traffic for one or more instances. Security groups consist of security group rule sets that define the allowed network traffic for all instances within the security groups.

[Load Balancers]

Load balancers distribute incoming network traffic across multiple server instances to improve availability and prevent overload. Configure load balancers to manage traffic flow, monitor health, and ensure efficient resource use.

[Domain Name System]

Manage domain name system (DNS) zones and records.

Supported Network Connections

Get an overview of supported network connections.

At the transfer point of the direct connection in the SAP data center, the following network ports are available for single-mode fiber optic connections:

- 10GBASE-LR: Supports 10 Gbps over long-range single-mode fiber.
- 100GBASE-LR4: Supports 100 Gbps over long-range single-mode fiber.

Other port types require a separate agreement.

Network Traffic Rules

Get an overview of network traffic that is always enabled, regardless of security groups applied.

Outbound Traffic

- ICMP IPv4/IPv6
- Metadata Service (HTTP @ 169.254.169.254:80)
- DHCPv4/v6 Requests

Inbound Traffic

- ICMP IPv4/IPv6
- DHCPv4/v6 Responses

External Networks

Projects need at least one external network for an Internet connection. Without these networks, users can't access instances in the private network, leading to isolation. Domain owners create and manage external networks and their associated subnets.

Related Information

[Creating Projects]

Private Networks

Private networks isolate resources within a project and assign internal IP addresses for communication. Create, edit, and delete private networks, manage subnets, and share private networks with other projects.

[Creating Private Networks](#)

Create isolated private networks to enable secure internal communication and resource management within your project.

[Editing Private Networks](#)

Update the name or admin state of an existing private network.

[Deleting Private Networks](#)

Permanently delete a private network you no longer need.

[Sharing Private Networks with other Projects](#)

Share private networks with other projects using access control.

[Creating Subnets in Private Networks](#)

Create subnets in private networks to segment traffic.

[Deleting Subnets from Private Networks](#)

Permanently delete subnets from private networks when they are no longer needed or part of the network's design. This action frees up IP address space and removes unused resources.

Creating Private Networks

Create isolated private networks to enable secure internal communication and resource management within your project.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the **Private Networks** tab page.
3. Choose **Create new**.
4. Enter a name.
5. Select the admin state: UP
6. Enter a subnet name.
7. Enter the network address and subnet mask (CIDR).

   → Recommendation

   Use the default CIDR range `10.180.0.0/16` or a subrange.

Related Information

[Creating Routers](#)

Editing Private Networks

Update the name or admin state of an existing private network.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. Select the admin state. You have the following options:
   - **UP**
   - **DOWN**
6. Choose **Save**.

Deleting Private Networks

Permanently delete a private network you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Delete**.

Sharing Private Networks with other Projects

Share private networks with other projects using access control.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You have a private network.

For more information, see [Creating Private Networks](#).

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Access Control**.

   A dialog box opens.
4. Choose +.
5. Select or enter the ID of the target project allowed to use the private network.
6. Choose **Add**.

Related Information

[Border Gateway Protocol VPN](#)

Creating Subnets in Private Networks

Create subnets in private networks to segment traffic.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

Context

Create subnets within predefined CIDR ranges, such as the default CIDR range `10.180.0.0/16` or its subranges.

> **i Note**
>
> To create subnet IP ranges outside these predefined ranges, use the CLI.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Manage Subnets**.

   A dialog box opens.
4. Choose +.
5. Enter a subnet name.
6. Enter the network address and subnet mask (CIDR).

   > **→ Recommendation**
   >
   > Use the default CIDR range `10.180.0.0/16` or a subrange.

7. Choose **Add**.

Related Information

[Command-Line Guide for SAP Cloud Infrastructure]

Deleting Subnets from Private Networks

Permanently delete subnets from private networks when they are no longer needed or part of the network's design. This action frees up IP address space and removes unused resources.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Manage Subnets**.

   A dialog box opens.
4. Choose 🗑.

Routers

A router connects subnets and external networks, enabling communication within your infrastructure. Manage router settings, configure routers, and view your router topology.

Routers Configuration Maximum

Get an overview of the configuration limits for routers in VM infrastructures.

Creating Routers

Create routers to connect multiple private network subnets to external networks. A router also enables one-to-one NAT between the private networks and external networks via floating IPs.

Editing Routers

Update the name, floating IP network, subnet, and private network subnets of an existing router.

Deleting Routers

Permanently delete a router you no longer need.

Viewing the Router Topology

Get a visual representation of the router topology.

Routers Configuration Maximum

Get an overview of the configuration limits for routers in VM infrastructures.

Resource	Maximum Size / Count	Comments
Private Networks per Router	100	Maximum number of private networks that can be connected to one router.
Floating IPs per Router	1024	Maximum number of floating IP addresses that can be assigned to a router.
Directly Accessible Private Network (DAPN) IPs / Ports per Router	1280	Maximum number of DAPN IPs/ports that can be passed through a router.
Static Routes per Router	256	Maximum number of all static IPv4/6 routes that can be assigned to a router. The router can dynamically learn additional routes through BGP VPN.

Creating Routers

Create routers to connect multiple private network subnets to external networks. A router also enables one-to-one NAT between the private networks and external networks via floating IPs.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

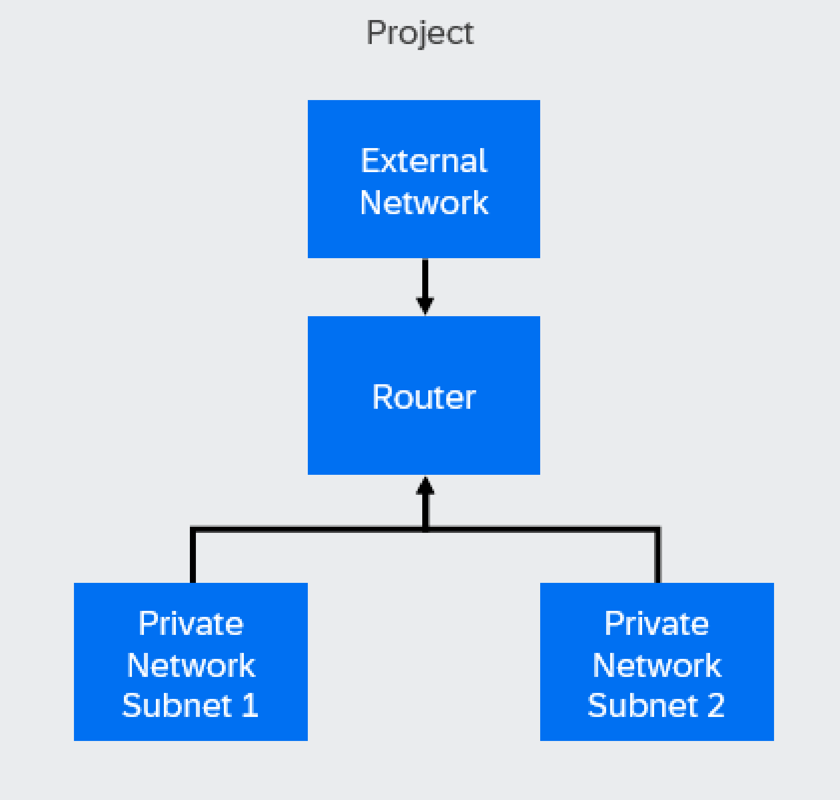
- You know and comply with the configuration limits for routers.

  For more information, see Routers Configuration Maximum.

- You have private networks.

  For more information, see Private Networks.

Context



A router connects multiple private network subnets to a single external network and enables one-to-one NAT for communication via floating IPs.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the Routers tab page.
3. Choose Create new.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Select the admin state. You have the following options:
   - UP
   - DOWN
6. Select the floating IP network.
7. Select the private network subnets to attach.

   You can attach private subnets to only one router at a time.
8. Choose Create.

Next Steps

To verify the router's configuration, try pinging from an instance in one private network subnet to an instance in another private network subnet.

Editing Routers

Update the name, floating IP network, subnet, and private network subnets of an existing router.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for routers.

  For more information, see Routers Configuration Maximum.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.
3. From the gear dropdown menu for the router, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. Select the floating IP network.
6. Select the subnet.
7. Select the private network subnets.
8. Choose **Update**.

Deleting Routers

Permanently delete a router you no longer need.

Prerequisites

- The router you want to delete has no associated objects, such as routes, gateways, or ports.
- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.
3. From the gear dropdown menu for the router, choose **Delete**.

Viewing the Router Topology

Get a visual representation of the router topology.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.

3. From the gear dropdown menu for the router, choose Topology.

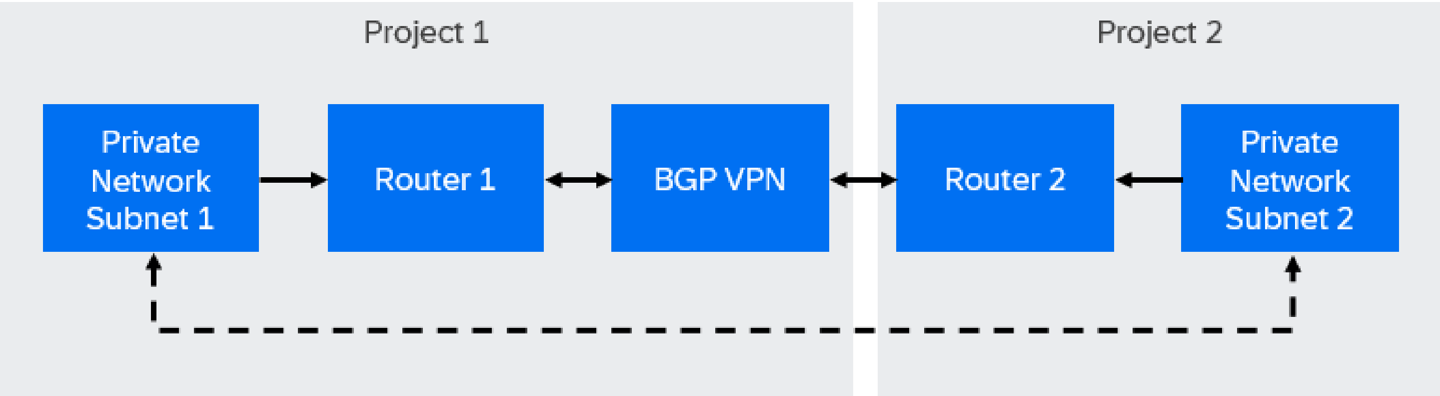
Border Gateway Protocol VPN

Border gateway protocol (BGP) VPNs are virtual private networks that use BGP to exchange routing information between connected networks. Manage BGP VPNs by creating them, adding or removing routers, sharing them with other projects, and deleting them when they are no longer needed.

Cross-Project Interconnections

Use border gateway protocol (BGP) VPN to enable communication between virtual machines (VMs) in different projects within the same region.

How It Works



Each private network is connected to a router, and the BGP VPN is created in one of the projects. The BGP VPN is then shared with the second project. By establishing associations between the routers and the BGP VPN, VMs in different projects can communicate directly without using public IP addresses.

Requirements

- Two separate private networks in different projects with distinct subnets.

  For more information, see:

  - Creating Private Networks
  - Creating Subnets in Private Networks

- A router connected to each private network.

  For more information, see Creating Routers.

- Both projects must belong to the same domain.

- A BGP VPN created in one of the projects.

  For more information, see Creating Border Gateway Protocol VPNs.

- The BGP VPN shared with the second project.

  For more information, see Sharing Border Gateway Protocol VPNs with other Projects.

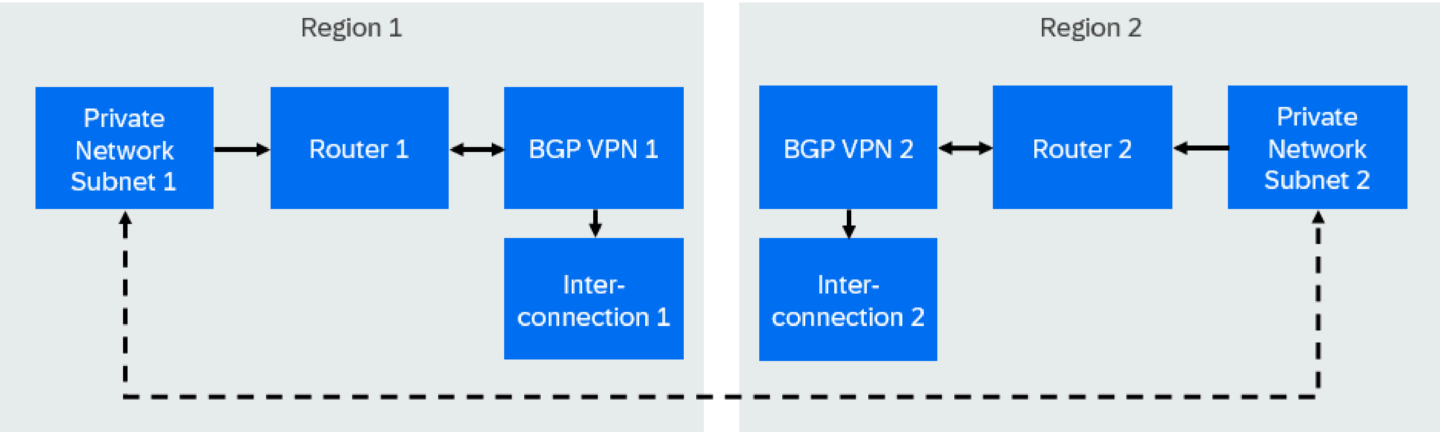
- Associations established between the routers and the BGP VPN.

  For more information, see Adding Routers to Border Gateway Protocol VPNs.

Cross-Region Interconnections

Use border gateway protocol (BGP) VPN to enable communication between networks located in different regions by creating interconnections between BGP VPNs.

How It Works



Each BGP VPN operates independently within its respective region. To establish communication between the private networks, an interconnection object is created, linking the BGP VPNs. This interconnection acts as a bridge, ensuring traffic from one region reaches the correct destination in the other region while maintaining isolation.

Requirements

- Private networks in each region with distinct, non-overlapping subnets.

  For more information, see:

  - Creating Private Networks

    - Creating Subnets in Private Networks

- A router connected to each private network.

  For more information, see Creating Routers.

- Both projects must belong to the same domain.

- Two separate BGP VPNs, one in each region.

  For more information, see Creating Border Gateway Protocol VPNs.

- An interconnection object that links both BGP VPNs.

  For more information, see Required Plugins.

Creating Border Gateway Protocol VPNs

Create border gateway protocol (BGP) VPNs for your project.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. Choose **New BGP VPN**.

   A dialog box opens.
4. Enter a name.
5. Choose **Save**.

Next Steps

- Adding Routers to Border Gateway Protocol VPNs
- Sharing Border Gateway Protocol VPNs with other Projects

Adding Routers to Border Gateway Protocol VPNs

Connect routers to border gateway protocol (BGP) VPNs to enable subnet announcements.

Prerequisites

- You have a router.

  For more information, see Creating Routers.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. From the gear dropdown menu for the BGP VPN, choose **Manage Routers**.

   A dialog box opens.

4. Select a router.
5. Choose Add.

Removing Routers from Border Gateway Protocol VPNs

Remove the association between routers and border gateway protocol (BGP) VPNs to stop announcing subnets connected to the routers.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the BGP VPNs tab page.
3. From the gear dropdown menu for the BGP VPN, choose Manage Routers.

   A dialog box opens.
4. Choose 🗑.
5. Choose Confirm.

Sharing Border Gateway Protocol VPNs with other Projects

Share border gateway protocol (BGP) VPNs with other projects using access control.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the BGP VPNs tab page.
3. From the gear dropdown menu for the BGP VPN, choose Access Control.

   A dialog box opens.
4. Enter the project name or ID of the target project.
5. Choose Add.

Deleting Border Gateway Protocol VPNs

Permanently delete a border gateway protocol (BGP) VPN you no longer need.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the BGP VPNs tab page.
3. From the gear dropdown menu for the BGP VPN, choose Delete.
4. Confirm the dialog box.

Fixed IPs / Ports

Fixed IPs are IPv4 addresses that DHCP assigns to server instances (VMs). A fixed IP acts as a port on the private network and connects to the NIC of a server instance.

[Reserving Fixed IPs](#)

Reserve fixed IP addresses to ensure consistent network connectivity for network devices or server instances (VMs) within your private network.

[Editing Fixed IP Ports](#)

Update the security group and description of an existing fixed IP port.

[Editing Ports](#)

Update the security group and description of an existing port.

[Deleting Fixed IP Ports](#)

Permanently delete a fixed IP port you no longer need.

Reserving Fixed IPs

Reserve fixed IP addresses to ensure consistent network connectivity for network devices or server instances (VMs) within your private network.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have created a security group.

  For more information, see Creating Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. Choose **Reserve new IP**.

   A dialog box opens.
3. Select the network.
4. Select the subnet.
5. **Optional:** Enter the IP.
6. **Optional:** Select the security group.
7. **Optional:** Enter a description.
8. Choose **Save**.

Related Information

Creating Server Instances

Editing Fixed IP Ports

Update the security group and description of an existing fixed IP port.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have created a security group.

  For more information, see Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. From the gear dropdown menu for the fixed IP port, choose **Edit**.

   A dialog box opens.
3. **Optional:** Select the security groups.
4. **Optional:** Enter the description.
5. Choose **Save**.

Editing Ports

Update the security group and description of an existing port.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. Select the **other ports** radio button.
3. From the gear dropdown menu for the port, choose **Edit**.

   A dialog box opens.
4. **Optional:** Select the security groups.
5. **Optional:** Enter the description.
6. Choose **Save**.

Deleting Fixed IP Ports

Permanently delete a fixed IP port you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. From the gear dropdown menu for the fixed IP port, choose **Delete**.
3. Confirm the dialog box.

Floating IPs

Floating IPs (FIPs) are static IPs that belong to a project rather than a specific server instance. You can assign them to any server instance for SSH, RDP, or HTTP access. FIPs are created from the IPv4 pool of a subnet.

Creating Floating IPs

Create a public IP address that you can later attach to your server instances to enable internet access. If you specify a DNS domain, the network service automatically creates an A record for name resolution.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

- You have an external network.

  For more information, see [External Networks](#).

- **Optional:** To automatically create an A record for the floating IP, you have a domain name system (DNS) zone.

  For more information, see [Domain Name System](#).

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. Choose **Allocate new**.

   A dialog box opens.
3. Select the external network.
4. Select the external subnet.
5. **Optional:** Select the DNS domain.
6. **Optional:** Enter the DNS name that you want to use with the floating IP.

Ensure that the DNS name is a valid partially qualified domain name (PQDN) or fully qualified domain name (FQDN) and is no longer than 63 characters.

7. Choose **Allocate**.

Results

- The network service allocates a floating IP and associates it with the selected external network.
- If you specified a DNS domain and name, the network service creates an A record for name resolution.
- The activity is recorded as an event in the audit log.

Next Steps

Attaching Floating IPs to Server Instances

Related Information

Audit

Editing Floating IPs

Update the description of an existing floating IP.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. From the gear dropdown menu for the floating IP, choose **Edit**.

   A dialog box opens.
3. Edit the description.
4. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Releasing Floating IPs

Release unused floating IPs back to the network to free up resources.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have detached the floating IP from any server instance.

  For more information, see Detaching Floating IPs from Server Instances.

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. From the gear dropdown menu for the floating IP, choose **Release**.

Results

- If the floating IP is attached to a server instance, it's automatically detached first. The released IP returns to the external network and becomes available for a new allocation.

- The activity is recorded as an event in the audit log.

Related Information

Creating Floating IPs
Audit

Security Groups

Manage security groups that act as virtual firewalls to control incoming (ingress) and outgoing (egress) network traffic for one or more instances. Security groups consist of security group rule sets that define the allowed network traffic for all instances within the security groups.

Security Groups Configuration Maximum
Get an overview of the configuration limits for security groups in VM infrastructures.

Creating Security Groups
Create security groups to manage network traffic for one or more instances.

Adding Security Group Rules to Security Groups
Security group rules specify the network traffic which is allowed for all instances within a security group.

Sharing Security Groups with other Projects
Define which target projects can use a security group. This enables reuse across multiple projects without duplicating settings and prevents target projects from modifying them.

Deleting Security Group Rules from Security Groups
Delete security rules that you no longer need from a security group.

Deleting Security Groups
Permanently delete a security group you no longer need.

Related Information

Network Traffic Rules

Security Groups Configuration Maximum

Get an overview of the configuration limits for security groups in VM infrastructures.

Resource	Maximum Size / Count
Security Group Rules per Security Group	1000
IPs or Subnets per Security Group Rule	4000
Security Groups per Port/Server	28

Creating Security Groups

Create security groups to manage network traffic for one or more instances.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for security groups.

  For more information, see Security Groups Configuration Maximum.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. Choose **Create New**.

   A dialog box opens.
3. Enter a name.
4. Enter a description.
5. Choose **Create**.

Results

The security group is created and automatically configured to allow all outgoing (egress) connections.

Next Steps

Adding Security Group Rules to Security Groups

Adding Security Group Rules to Security Groups

Security group rules specify the network traffic which is allowed for all instances within a security group.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for security groups.

  For more information, see Security Groups Configuration Maximum.

- You know the network traffic rules.

  For more information, see Network Traffic Rules.

Context

A common use case for adding security group rules is to allow remote access to server instances by creating an ingress rule for SSH (port 22) or RDP (port 3389).

Procedure

1. Choose Networking & Loadbalancing > Security Groups.
2. Choose the security group.
3. Choose **Add New Rule**.
4. Select the rule type.
5. Select the protocol.
6. Select the direction. You have the following options:
    - **Ingress**

      An ingress rule defines which origin networks are allowed to connect to the instance and on which ports.

    - **Egress**

      An egress rule defines which target networks the instance is allowed to connect to.
7. Enter a specific port, such as 80, or a port range, such as 0-80.

   SAP Cloud Infrastructure enforces port security for all supported ports.
8. Select the remote source of the network traffic. You have the following options:
    - **IP Address**
    - **Security Group**
9. Enter the IP address, IP range, or security group.

   ### Example

    - Specific IP address: `10.180.1.1/32`
    - IP range: `10.180.1.0/24`

   #### Caution

   `0.0.0.0/0` allows network traffic from or to any IP address.
10. Choose **Create**.

Sharing Security Groups with other Projects

Define which target projects can use a security group. This enables reuse across multiple projects without duplicating settings and prevents target projects from modifying them.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You have a security group other than the **default** security group.

  Note

  You cannot share the **default** security group.

  For more information, see Creating Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. From the gear dropdown menu for the security group, choose **Access Control**.

   A dialog box opens.
3. Enter the project name or ID of the target project.
4. Choose **Add**.

Deleting Security Group Rules from Security Groups

Delete security rules that you no longer need from a security group.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. Choose the security group.
3. Choose the delete button next to the security rule you want to delete.
4. Confirm the dialog box.

Deleting Security Groups

Permanently delete a security group you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Context

Ensure that you delete any unused security groups that aren't attached to any instances. Deleting unused security groups not only keeps your environment clean but also ensures that you don't accidentally attach them to any instances, which could inadvertently open up your environment to attacks.

Note
You can't delete the **default** security group.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. From the gear dropdown menu for the security group, choose **Delete**.
3. Confirm the dialog box.

Load Balancers

Load balancers distribute incoming network traffic across multiple server instances to improve availability and prevent overload. Configure load balancers to manage traffic flow, monitor health, and ensure efficient resource use.

Load Balancer Statuses
Get an overview of the possible statuses for load balancers, listeners, pools, pool members, and health monitors.

Creating Load Balancers
Create load balancers to automatically distribute network traffic across multiple server instances.

Editing Load Balancers
Update the name, description, and tags of an existing load balancer.

Deleting Load Balancers
Permanently delete a load balancer you no longer need.

Viewing Load Balancer JSON Configuration
View the JSON configuration of the load balancer.

Adding Floating IPs to Load Balancers
Add a floating IP to a load balancer.

Removing Floating IPs from Load Balancers
Remove a floating IP that is no longer needed from a load balancer.

Listeners
Listeners define how the load balancer processes incoming traffic by specifying protocols and ports. Create and configure listeners, and apply traffic management policies to control how traffic is directed to backend resources.

Pools
Pools define backend server groups for traffic distribution. Configure and manage pool members, monitor health, and adjust settings.

Load Balancer Statuses

Get an overview of the possible statuses for load balancers, listeners, pools, pool members, and health monitors.

Note

Status updates are subject to the polling interval. After changing load balancer objects, allow up to 2 minutes for the status to update.

Load Balancer Statuses

Load Balancer Status	Description	API Status Equivalent
ONLINE	The load balancer is operational and serving traffic.	ACTIVE
OFFLINE	The load balancer is not serving traffic, possibly due to unhealthy amphorae, errors, or all pool members being unhealthy.	ERROR, non-operational state
PENDING	The load balancer is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE
DEGRADED	Some objects in the hierarchy (such as pool members) are unhealthy (for example, OFFLINE), but the load balancer is still operational.	ACTIVE (with unhealthy components)

Listener Statuses

Listener Status	Description	API Status Equivalent
ONLINE	The listener is configured and ready to accept traffic, but requires an operational load balancer.	ACTIVE
OFFLINE	The listener is not operational, possibly due to errors or being disabled.	ERROR, non-operational state
PENDING	The listener is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE
DEGRADED	The listener has reduced functionality, typically during failover in Active/Standby topology (context-specific).	

Pool Statuses

Pool Status	Description	API Status Equivalent
ONLINE	The pool is operational with at least some healthy members.	ACTIVE
OFFLINE	The pool is not operational, often due to all members being unhealthy or errors.	ERROR
DEGRADED	Some members are unhealthy (for example, OFFLINE), but the pool is still operational.	ACTIVE (with unhealthy members)
PENDING	The pool is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE

Pool Member Statuses

Pool Member Status	Description	API Status Equivalent
ONLINE	The member is healthy and receiving traffic.	ONLINE
OFFLINE	The member is not receiving traffic due to configuration or health check failure.	OFFLINE
ERROR	The member has failed health checks.	ERROR
DRAINING	The member is being gracefully removed, allowing existing connections to complete.	DRAINING
NO_MONITOR	No health monitor is configured, so health is unknown.	NO_MONITOR

Health Monitor Statuses

Health Monitor Status	Description	API Status Equivalent
ONLINE	The health monitor is actively checking member health.	ONLINE
OFFLINE	The health monitor is disabled or not running.	OFFLINE
ERROR	An error occurred during monitoring.	ERROR

Creating Load Balancers

Create load balancers to automatically distribute network traffic across multiple server instances.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have sufficient quota. To create a load balancer, you need quota for one to three ports: one for the virtual IP (VIP) and two for SelfIP addresses. The two SelfIP addresses are only required if the load balancer is the first one in its subnet.

  For more information, see Resource Management.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose **New Load Balancer**.

   A dialog box opens.
3. Enter a name.
4. **Optional:** Enter a description.
5. Select a private network.
6. **Optional:** To configure advanced network options, choose **Toggle advanced network options**.
   a. Select the subnet for the load balancer.
   b. To automatically allocate a new IP address from the private network, leave the IP address field empty, or enter a specific IP address.
7. **Optional:** Enter tags to organize or classify the load balancer.
8. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Listeners
- Pools

Related Information

Audit

Editing Load Balancers

Update the name, description, and tags of an existing load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. From the gear dropdown menu for the load balancer, choose **Edit**.

A dialog box opens.

3. Enter a name.
4. **Optional:** Enter a description.
5. **Optional:** Enter tags to organize or classify the load balancer.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Load Balancers

Permanently delete a load balancer you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. From the gear dropdown menu for the load balancer, choose **Delete**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Load Balancer JSON Configuration

View the JSON configuration of the load balancer.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers .
2. From the gear dropdown menu for the load balancer, choose JSON.

   A dialog box opens and shows the load balancer JSON configuration.

Adding Floating IPs to Load Balancers

Add a floating IP to a load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a floating IP.

  For more information, see Floating IPs.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers .
2. From the gear dropdown menu for the load balancer, choose Attach Floating IP.

   A dialog box opens.
3. Select the floating IP.
4. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Removing Floating IPs from Load Balancers

Remove a floating IP that is no longer needed from a load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers .

2. From the gear dropdown menu for the load balancer, choose **Detach Floating IP**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Listeners

Listeners define how the load balancer processes incoming traffic by specifying protocols and ports. Create and configure listeners, and apply traffic management policies to control how traffic is directed to backend resources.

Extended Policies

Get an overview of the extended policies available for your load balancer. These policies allow you to fine-tune how traffic is managed and processed, from enabling compression, encryption, and proxy protocols to controlling header overrides and protocol settings.

TLS Cipher Suite Catalog

Get an overview of the default cipher suites attached to listeners using TLS-terminated HTTPS protocol. Each cipher suite defines specific encryption algorithms and protocols used for securing network traffic.

HTTP and TLS-Terminated HTTPS Listener Headers

Get an overview of the optional headers you can insert into HTTP and TLS-terminated HTTPS listeners. These headers provide additional information about the incoming requests, which can be forwarded to the pool members.

Creating HTTP Listeners

Add a listener for the hypertext transfer protocol (HTTP) to forward HTTP-based traffic to the backend pool.

Creating TCP Listeners

Add a listener for the Transmission Control Protocol (TCP) to forward raw TCP traffic to a backend pool.

Creating TLS-Terminated HTTPS Listeners

Add a listener for the TLS-terminated hypertext transfer protocol (HTTPS) to handle encrypted HTTPS traffic, decrypt it at the load balancer, and forward it to the backend pool.

Creating UDP Listeners

Add a User Datagram Protocol (UDP) listener to forward UDP-based traffic (for example, DNS) to the backend pool.

Editing Listeners

Update the name, description, default pool, or connection limit of an existing listener.

Deleting Listeners

Delete listeners that you no longer need.

Configuring Allowed CIDRs for Listeners

Restrict traffic by filtering source IP addresses and subnets.

Viewing Listener JSON Configuration

View the JSON configuration of the listener.

Creating L7 Policies

Create L7 policies to block or reroute traffic based on L7 rules.

Editing L7 Policies

Update the name, description, position, action, or tags of an existing L7 policy.

Deleting L7 Policies

Delete L7 policies that you no longer need.

Viewing L7 Policy JSON Configuration

View the JSON configuration of the L7 policy.

[Creating L7 Rules]

Create L7 rules to manage how the load balancer handles application-layer traffic, such as HTTP requests, based on specific request characteristics. These rules are part of an L7 policy, and all rules in a policy are logically ANDed together.

[Editing L7 Rules]

Update the rule type, comparison type, value, and tags of an existing L7 rule.

[Deleting L7 Rules]

Delete L7 rules that you no longer need.

[Viewing L7 Rule JSON Configuration]

View the JSON configuration of the L7 rule.

Extended Policies

Get an overview of the extended policies available for your load balancer. These policies allow you to fine-tune how traffic is managed and processed, from enabling compression, encryption, and proxy protocols to controlling header overrides and protocol settings.

Extended Policy	Tag	Protocol Type	Description
Enable HTTP Compression	http_compression_e4a2_v1_0	HTTP TERMINATED_HTTPS	Apply a compression profile to your listener to compress all content of type text/* and application/(xml|x-javascript) using the gzip method. Make sure your content is at least 1024 bytes long to enable compression.
Enable Cookie Encryption	cookie_encryption_b82a_v1_0	HTTP TERMINATED_HTTPS	Encrypt all cookies that are sent to your client and automatically decrypt them for the pool members.
Enable Proxy Protocol v1	proxy_protocol_2edF_v1_0	TCP HTTPS HTTP TERMINATED_HTTPS	Enable version 1 of the proxy protocol for your listener. Ensure your pool members support this protocol to avoid backend errors. You can also apply this setting to listeners with protocol TCP if you have the tag standard_tcp_a3de_v1_0 set.
Enable Proxy Protocol v2	proxy_protocol_V2_e8f6_v1_0	TCP HTTPS HTTP TERMINATED_HTTPS	Enable version 2 of the proxy protocol for your listener. Make sure your pool members support this protocol to prevent errors on the backend. You can also apply this to listeners with protocol TCP if the tag standard_tcp_a3de_v1_0 is set.
Disable HW Acceleration	standard_tcp_a3de_v1_0	TCP	Disable hardware acceleration for listeners using the TCP or

Extended Policy	Tag	Protocol Type	Description
		HTTPS	HTTPS protocol. Disabling is required, for example, when you need to use the proxy protocol (either version).
Enable HTTP to HTTPS Redirect	http_redirect_a26c_v1_0	HTTP	Automatically redirect all HTTP traffic to HTTPS to secure your connections.
Enable Stateless UDP Load Balancing	ccloud_special_udp_stateless	UDP	By default, your load balancer sends every UDP packet from a client to the same pool member, which is helpful for longer client-server communications. Enable stateless UDP load balancing to distribute UDP packets across multiple pool members and improve scalability and performance.
Disable SNAT	ccloud_special_l4_deactivate_snat	TCP HTTPS HTTP TERMINATED_HTTPS UDP	By default, your load balancer uses source network address translation (SNAT). Use this setting to disable SNAT, so your pool member can see the original IP address of the client. **Note** When you disable SNAT, make sure your pool members have routes that return traffic to the load balancer IP address. Without this, TCP connections will fail, as the initiator expects ACK packets from the load balancer IP, not directly from the pool member.
Override X-Forwarded-Host Header	ccloud_special_xfh_override	HTTP TERMINATED_HTTPS	Ensure full control over the X-Forwarded-Host header. If this header is present in an incoming HTTP(S) request, the load balancer overrides its value with the value from the built-in HTTP Host header. This helps you prevent malicious header injection by clients.
Enable TCP Connection Mirroring	ccloud_special_tcp_mirror	TCP	Enable mirroring of TCP connections between active and standby devices in the load balancer infrastructure. This helps maintain your TCP connections without

Extended Policy	Tag	Protocol Type	Description
			interruption during maintenance, failovers, or reboots. Use this option if you need high resiliency for long-lived TCP connections. **Note** TCP connection mirroring is only supported for listeners with the TCP protocol and is not suitable for short-lived connections like typical HTTP(S) traffic.

Related Information

Creating HTTP Listeners
Creating TCP Listeners
Creating TLS-Terminated HTTPS Listeners
Creating UDP Listeners

TLS Cipher Suite Catalog

Get an overview of the default cipher suites attached to listeners using TLS-terminated HTTPS protocol. Each cipher suite defines specific encryption algorithms and protocols used for securing network traffic.

Cipher Suite	Cipher Type	Description
AES128-GCM-SHA256	AES	AES128 encryption in GCM mode with SHA-256 for message authentication
AES128-SHA	AES	AES128 encryption with SHA for message authentication
AES128-SHA256	AES	AES128 encryption with SHA-256 for message authentication
AES256-GCM-SHA384	AES	AES256 encryption in GCM mode with SHA-384 for message authentication
AES256-SHA	AES	AES256 encryption with SHA for message authentication
AES256-SHA256	AES	AES256 encryption with SHA-256 for message authentication
CAMELLIA128-SHA	Camellia	Camellia128 encryption with SHA for message authentication
CAMELLIA256-SHA	Camellia	Camellia256 encryption with SHA for message authentication
ECDHE-ECDSA-AES128-GCM-SHA256	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption in GCM

Cipher Suite	Cipher Type	Description
		mode, and SHA-256 for message authentication
ECDHE-ECDSA-AES128-SHA	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption, and SHA for message authentication
ECDHE-ECDSA-AES128-SHA256	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption, and SHA-256 for message authentication
ECDHE-ECDSA-AES256-GCM-SHA384	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption in GCM mode, and SHA-384 for message authentication
ECDHE-ECDSA-AES256-SHA	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption, and SHA for message authentication
ECDHE-ECDSA-AES256-SHA384	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption, and SHA-384 for message authentication
ECDHE-RSA-AES128-CBC-SHA	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES128 encryption in CBC mode, and SHA for message authentication
ECDHE-RSA-AES128-GCM-SHA256	ECDHE-ECDSA	ECDHE key exchange with RSA authentication, AES128 encryption in GCM mode, and SHA-256 for message authentication
ECDHE-RSA-AES128-SHA256	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES128 encryption, and SHA-256 for message authentication
ECDHE-RSA-AES256-CBC-SHA	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption in CBC mode, and SHA for message authentication
ECDHE-RSA-AES256-GCM-SHA384	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption in GCM mode, and SHA-384 for message authentication
ECDHE-RSA-AES256-SHA384	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption, and SHA-384 for message authentication
ECDHE-RSA-CHACHA20-POLY1305-SHA256	ECDHE-RSA	ECDHE key exchange with RSA authentication, ChaCha20 encryption, and Poly1305 for message authentication
TLS13-AES128-GCM-SHA256	TLS 1.3	TLS 1.3 cipher suite with AES128 encryption in GCM mode and SHA-256 for message authentication

Cipher Suite	Cipher Type	Description
TLS13-AES256-GCM-SHA384	TLS 1.3	TLS 1.3 cipher suite with AES256 encryption in GCM mode and SHA-384 for message authentication
TLS13-CHACHA20-POLY1305-SHA256	TLS 1.3	TLS 1.3 cipher suite with ChaCha20 encryption and Poly1305 for message authentication

HTTP and TLS-Terminated HTTPS Listener Headers

Get an overview of the optional headers you can insert into HTTP and TLS-terminated HTTPS listeners. These headers provide additional information about the incoming requests, which can be forwarded to the pool members.

Header	Description	Protocol Type
X-Forwarded-For	Adds the original client IP address.	HTTP TERMINATED_HTTPS
X-Forwarded-Port	Adds the original port number used by the client.	HTTP TERMINATED_HTTPS
X-Forwarded-Proto	Adds the original protocol (HTTP or HTTPS).	HTTP TERMINATED_HTTPS
X-SSL-Client-Verify	Indicates the verification status of the client certificate.	TERMINATED_HTTPS
X-SSL-Has-Cert	Indicates whether the client certificate is present.	TERMINATED_HTTPS
X-SSL-Client-DN	Adds the distinguished name (DN) of the client certificate.	TERMINATED_HTTPS
X-SSL-Client-CN	Adds the common name (CN) of the client certificate.	TERMINATED_HTTPS
X-SSL-Issuer	Adds the issuer of the client certificate.	TERMINATED_HTTPS
X-SSL-ClientSHA1	Adds the SHA1 fingerprint of the client certificate.	TERMINATED_HTTPS
X-SSL-Client-Not-Before	Adds the "Not Before" timestamp from the client certificate.	TERMINATED_HTTPS
X-SSL-Client-Not-After	Adds the "Not After" timestamp from the client certificate.	TERMINATED_HTTPS

Creating HTTP Listeners

Add a listener for the hypertext transfer protocol (HTTP) to forward HTTP-based traffic to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

- **Optional:** If you want to insert headers into the request before it is sent to the pool member, ensure you are familiar with the available headers and their meaning.

  For more information, see HTTP and TLS-Terminated HTTPS Listener Headers.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **HTTP**
8. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy will always apply, and L7 rules are not applicable.

   The extended policy is displayed as a tag.
9. **Optional:** Insert headers into the request before it is sent to the pool member.
10. **Optional:** Select a default pool.

    All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
11. **Optional:** Adjust the connection limit.
12. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
13. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating L7 Policies

Related Information

Audit

Creating TCP Listeners

Add a listener for the Transmission Control Protocol (TCP) to forward raw TCP traffic to a backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **TCP**
8. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy will always apply, and L7 rules are not applicable.

   The extended policy is displayed as a tag.
9. **Optional:** Select a default pool.

   All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
10. **Optional:** Adjust the connection limit.
11. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
12. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Creating TLS-Terminated HTTPS Listeners

Add a listener for the TLS-terminated hypertext transfer protocol (HTTPS) to handle encrypted HTTPS traffic, decrypt it at the load balancer, and forward it to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- You have a secret containing a PKCS12 format certificate/key.

  For more information, see Key Manager.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

- **Optional:** If you want to insert headers into the request before it is sent to the pool member, ensure you are familiar with the available headers and their meaning.

  For more information, see HTTP and TLS-Terminated HTTPS Listener Headers.

- **Optional:** If you want to change the default cipher suites attached to listeners, make sure you are familiar with the cipher suites and understand their implications.

  For more information, see TLS Cipher Suite Catalog.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **TERMINATED_HTTPS**
8. Select the secret containing a PKCS12 format certificate/key.
9. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy is displayed as a tag.
10. **Optional:** Insert headers into the request before it is sent to the pool member.
11. **Optional:** When you have multiple TLS certificates that you would like to use on the same listener, select the server name indication (SNI) secrets.
12. **Optional:** Select the client authentication mode. You have the following options:
    - **NONE**
    - **OPTIONAL**
    - **MANDATORY**

13. **Optional:** Select the client authentication secret.
14. **Optional:** Remove default TLS ciphers suites attached to the listener.
15. **Optional:** Select a default pool.

    All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
16. **Optional:** Adjust the connection limit.
17. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
18. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating L7 Policies

Related Information

Audit

Creating UDP Listeners

Add a User Datagram Protocol (UDP) listener to forward UDP-based traffic (for example, DNS) to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.
- You have a load balancer.
- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **UDP**

8. **Optional:** Select a default pool.

   All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.

9. **Optional:** Adjust the connection limit.
10. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
11. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing Listeners

Update the name, description, default pool, or connection limit of an existing listener.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has listeners.
3. From the gear dropdown menu for the listener, choose **Edit**.

   A dialog box opens.

4. Update any fields.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Listeners

Delete listeners that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has listeners.
3. From the gear dropdown menu for the listener, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Configuring Allowed CIDRs for Listeners

Restrict traffic by filtering source IP addresses and subnets.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Context

By default, all traffic is allowed. If you specify a list of CIDRs, the default behavior changes to deny all traffic except for the specified CIDRs.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that contains the listener you want to configure.
3. From the gear dropdown menu for the listener, choose **Allowed CIDRs**.

   A dialog box opens.
4. Enter one or more IPv4 or IPv6 CIDRs.
5. Choose **Save**.

Results

- Once a list of CIDRs is provided, all other traffic is denied by default.

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Listener JSON Configuration

View the JSON configuration of the listener.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has listeners.
3. From the gear dropdown menu for the listener, choose **JSON**.

   A dialog box opens and shows the listener JSON configuration.

Creating L7 Policies

Create L7 policies to block or reroute traffic based on L7 rules.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

  For more information, see Creating Load Balancers.

- You have an HTTP or TLS-terminated HTTPS listener.

  For more information, see:

  - Creating HTTP Listeners
  - Creating TLS-Terminated HTTPS Listeners

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer containing the listener to which you want to add a L7 policy.
3. Choose the listener.
4. Choose **New L7 Policy**.

   A dialog box opens.

5. Enter a name.
6. **Optional:** Enter a description.
7. **Optional:** Change the position of the policy.

   L7 policies are evaluated in the order defined by the position attribute. The first policy that matches a request determines the action taken. If no policy matches, the request routes to the listener's default pool, if it exists.
8. Select an action that will be executed when all L7 policies are matched. You have the following options:

Action	Description
REDIRECT_PREFIX	Redirect requests to a specified prefix. Select the redirect HTTP code (301, 302, 303, 307, or 308) and enter the redirect prefix.
REDIRECT_TO_POOL	Redirect requests to a specific pool. Enter the pool ID.
REDIRECT_TO_URL	Redirect requests to a specific URL. Select the redirect HTTP code (301, 302, 303, 307, or 308) and the URL.
REJECT	Reject the request.

9. **Optional:** Enter tags to organize or classify the L7 policy.
10. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Creating L7 Rules

Related Information

Audit

Editing L7 Policies

Update the name, description, position, action, or tags of an existing L7 policy.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers .
2. Choose the load balancer.
3. Choose the listener.
4. From the gear dropdown menu for the policy, choose **Edit**.

   A dialog box opens.
5. Enter a name.
6. **Optional:** Enter a description.
7. **Optional:** Change the position of the policy.

   L7 policies are evaluated in the order defined by the position attribute. The first policy that matches a request determines the action taken. If no policy matches, the request routes to the listener's default pool, if it exists.
8. Select an action that will be executed when all L7 policies are matched. You have the following options:

Action	Description
REDIRECT_PREFIX	Redirect requests to a specified prefix. Select the redirect HTTP code (301, 302, 303, 307, or 308) and enter the redirect prefix.
REDIRECT_TO_POOL	Redirect requests to a specific pool. Enter the pool ID.
REDIRECT_TO_URL	Redirect requests to a specific URL. Select the redirect HTTP code (301, 302, 303, 307, or 308) and the URL.
REJECT	Reject the request.

9. **Optional:** Enter tags to organize or classify the L7 policy.
10. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting L7 Policies

Delete L7 policies that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have deleted all L7 rules from the L7 policy.

  For more information, see Deleting L7 Rules.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer containing the listener from which you want to delete a L7 policy.
3. Choose the listener.
4. From the gear dropdown menu for the L7 policy, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing L7 Policy JSON Configuration

View the JSON configuration of the L7 policy.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the listener.
4. From the gear dropdown menu for the L7 policy, choose **JSON**.

   A dialog box opens and shows the L7 policy JSON configuration.

Creating L7 Rules

Create L7 rules to manage how the load balancer handles application-layer traffic, such as HTTP requests, based on specific request characteristics. These rules are part of an L7 policy, and all rules in a policy are logically ANDed together.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have an L7 policy.

  For more information, see Creating L7 Policies.

Context

Without L7 rules, requests route to the listener's default pool. L7 rules let the listener evaluate requests and make routing decisions based on specific criteria. This eliminates the need for multiple load balancers.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. Choose **New L7 Rule**.

   A dialog box opens.
6. Select the rule type. You have the following options:

Rule Type	Description
COOKIE	Matches against a cookie name or value.
FILE_TYPE	Matches based on the file extension in the request URI.
HEADER	Matches the value of a specific HTTP header.
HOST_NAME	Matches the hostname from the HTTP request.
PATH	Matches the request URI path.
SSL_CONN_HAS_CERT	Matches whether the SSL connection presents a client certificate.
SSL_VERIFY_RESULT	Matches the result of the SSL certificate verification.
SSL_DN_FIELD	Matches fields in the SSL certificate's Distinguished Name (DN), such as CN or O.

7. Select the compare type. You have the following options:

Compare Type	Description
CONTAINS	Checks if the request field includes the specified value.
ENDS_WITH	Checks if the request field ends with the specified value.
EQUAL_TO	Checks if the request field exactly matches the specified value.
STARTS_WITH	Checks if the request field starts with the specified value.

8. **Optional:** To invert the logic of the rule, select the **Inverse Comparisation (NOT)** checkbox.

Example

If you select the comparison type EQUAL_TO and select inverse comparison, the rule matches when the values are not equal.

9. Enter the value to compare against, based on the selected rule type.
10. **Optional:** Enter tags to organize or classify the rule.
11. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing L7 Rules

Update the rule type, comparison type, value, and tags of an existing L7 rule.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the rule, choose Edit.

   A dialog box opens.
6. Select the rule type. You have the following options:

Rule Type	Description
COOKIE	Matches against a cookie name or value.
FILE_TYPE	Matches based on the file extension in the request URI.
HEADER	Matches the value of a specific HTTP header.
HOST_NAME	Matches the hostname from the HTTP request.
PATH	Matches the request URI path.
SSL_CONN_HAS_CERT	Matches whether the SSL connection presents a client certificate.

Rule Type	Description
SSL_VERIFY_RESULT	Matches the result of the SSL certificate verification.
SSL_DN_FIELD	Matches fields in the SSL certificate's Distinguished Name (DN), such as CN or O.

7. Select the compare type. You have the following options:

Compare Type	Description
CONTAINS	Checks if the request field includes the specified value.
ENDS_WITH	Checks if the request field ends with the specified value.
EQUAL_TO	Checks if the request field exactly matches the specified value.
STARTS_WITH	Checks if the request field starts with the specified value.

8. **Optional:** To invert the logic of the rule, select the Inverse Comparisation (NOT) checkbox.

   Example

   If you select the comparison type EQUAL_TO and select inverse comparison, the rule matches when the values are not equal.

9. Enter the value to compare against, based on the selected rule type.
10. **Optional:** Enter tags to organize or classify the rule.
11. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting L7 Rules

Delete L7 rules that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.

3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the L7 rule, choose **Delete**.
6. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing L7 Rule JSON Configuration

View the JSON configuration of the L7 rule.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the L7 rule, choose **JSON**.

   A dialog box opens and shows the L7 rule JSON configuration.

Pools

Pools define backend server groups for traffic distribution. Configure and manage pool members, monitor health, and adjust settings.

[Creating Pools]

Create pools to define which pool members (your backend servers) the load balancer sends traffic to. You also select how the load balancer distributes traffic between these pool members.

[Editing Pools]

Update the name, description, algorithm, session persistence type, or tags of an existing pool.

[Deleting Pools]

Permanently delete a pool you no longer need.

[Viewing Pool JSON Configuration]

View the JSON configuration of the pool.

Creating Pool Members

Create a new pool member that serves the traffic behind a load balancer. Each pool member is indicated by the IP address and port that it uses for traffic.

Editing Pool Members

Update the name, IPs, weight, or tags of an existing pool member.

Deleting Pool Members

Delete pool members that you no longer need.

Viewing Pool Member JSON Configuration

View the JSON configuration of the pool member.

Creating TCP, PING, TLS-HELLO, and UDP Health Monitors

Create a health monitor that uses low-level network probes to check whether pool members respond to connection attempts.

Creating HTTP and HTTPS Health Monitors

Create a health monitor that uses HTTP or HTTPS probes to check the availability of pool members based on response codes and path checks.

Editing Health Monitors

Update the name, max retries, or interval of an existing health monitor.

Deleting Health Monitors

Permanently delete a health monitor you no longer need.

Viewing Health Monitor JSON Configuration

View the JSON configuration of the health monitor.

Creating Pools

Create pools to define which pool members (your backend servers) the load balancer sends traffic to. You also select how the load balancer distributes traffic between these pool members.

Prerequisites

- You have a load balancer.
- **Optional:** If you want to encrypt connections to pool members with TLS, ensure the following:
  - You have created a certificate secret for TLS encryption and an authentication secret (CA) for validating connections to pool members.

    For more information, see Key Manager.
  - To customize the default cipher suites, make sure you understand how cipher suites work and their impact on security.

    For more information, see TLS Cipher Suite Catalog.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers .
2. Choose the load balancer to which you want to add a pool.
3. Choose **New Pool**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.

6. To define how the load balancer distributes traffic across the pool members, select a load balancing algorithm. You have the following options:

Load Balancing Algorithm	Description
LEAST_CONNECTIONS	The load balancer tracks active connections and forwards new requests to the pool member with the fewest open connections.
ROUND_ROBIN	The load balancer forwards each new request to the next pool member in line, distributing traffic evenly across all pool members. Use this method if your pool members have similar capacity and performance.

7. Select the protocol for routing traffic to pool members. You have the following options:
    - HTTP
    - PROXY
    - TCP
    - UDP
8. **Optional:** To ensure that traffic from the same client consistently reaches the same pool member, select a session persistence type.

    For each listener, you can only configure session persistence on its default pool. If you use other pools (for example, with an L7 policy), they follow the persistence settings of the default pool. You have the following options:

Session Persistence Type	Description
SOURCE_IP	The load balancer uses the client's IP address to keep all requests from the same client directed to the same pool member.
HTTP_COOKIE	The load balancer creates a cookie when it receives the first client request and uses it to route all subsequent requests from the same client to the same pool member.
APP_COOKIE	The load balancer uses a cookie created by the backend application to route all requests carrying the same cookie value to the same pool member.

9. **Optional:** To set this pool as the default for a listener, select the listener.
10. **Optional:** To encrypt connections to pool members with TLS, select the **Use TLS** checkbox.

    You can only attach a TLS-enabled pool to a TLS-terminated HTTPS listener.

    a. Select the certificate secret.

    b. Select the authentication secret (CA).

    c. Adjust the default TLS cipher suites attached to the listener.
11. **Optional:** Enter tags to organize or classify the pool.
12. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Creating TCP, PING, TLS-HELLO, and UDP Health Monitors
- Creating HTTP and HTTPS Health Monitors
- Creating Pool Members

Related Information

Audit

Editing Pools

Update the name, description, algorithm, session persistence type, or tags of an existing pool.

Prerequisites

- **Optional:** If you want to encrypt connections to pool members with TLS, ensure the following:
  - You have created a certificate secret for TLS encryption and an authentication secret (CA) for validating connections to pool members.

    For more information, see Key Manager.
  - To customize the default cipher suites, make sure you understand how cipher suites work and their impact on security.

    For more information, see TLS Cipher Suite Catalog.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that contains the pool you want to edit.
3. From the gear dropdown menu for the pool, choose **Edit**.

   A dialog box opens.
4. Update the fields you want to change.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Pools

Permanently delete a pool you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that contains the pool you want to delete.
3. From the gear dropdown menu for the pool, choose **Delete**.
4. Confirm the dialog box.

Results

- The pool is deleted, along with all associated pool members and health monitors, if any.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Pool JSON Configuration

View the JSON configuration of the pool.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has pool.
3. From the gear dropdown menu for the pool, choose **JSON**.

   A dialog box opens and shows the pool JSON configuration.

Creating Pool Members

Create a new pool member that serves the traffic behind a load balancer. Each pool member is indicated by the IP address and port that it uses for traffic.

Prerequisites

- You have a load balancer.

  For more information, see Creating Load Balancers.
- You have a pool.

  For more information, see Creating Pools.
- You have security groups that allow dynamic health checks.

  Load balancers undergo scaling or migration, which results in changes to health-check IPs. To avoid service interruptions, ensure that your security groups for pool members allow access from the full subnet of the load balancer.

Example

If your load balancer uses the subnet 10.180.1.0/24 and your pool members listen on port 1080, the security group has a TCP-INGRESS rule allowing 10.180.1.0/24 access to port 1080.

For more information, see Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. Choose **New Member**.

   A dialog box opens.
5. Enter a name.
6. Enter the IPs (IP address and protocol port) of the server where you want to route traffic.

   Do not add a pool member with the IP address of another load balancer in the same network.

   ### Remember

   If you add an IP from an existing server as a pool member and later change its IP address, the new IP will not be updated automatically. To update the IP, you must remove the pool member and then add it again.
7. **Optional:** Set the pool member as backup member.

   Backup members will not receive traffic unless all other pool members are down.
8. **Optional:** Adjust the weight for the new pool member.

   A weight between 0 and 100 determines how much traffic the pool member will handle. The default weight is 1.

   - To deactivate the pool member, set the weight to 0 (no traffic will be routed to it).
   - To increase the traffic for the pool member, enter a weight between 1 and 100. Higher values will result in more traffic for the member.

     ### Note

     Weights are uniformly mapped from the 0–255 range to the 0–100 range in the backend. This means that, for example, pool members with a weight of 1 and 2 will both be mapped to a weight of 1.
9. **Optional:** Enter tags to organize or classify the pool member.
10. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing Pool Members

Update the name, IPs, weight, or tags of an existing pool member.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menur for the pool member, choose **Edit**.
5. Enter a name.
6. Enter the IPs (IP address and protocol port) of the server where you want to route traffic.

   Do not add a pool member with the IP address of another load balancer in the same network.

   > **Remember**
   >
   > If you add an IP from an existing server as a pool member and later change its IP address, the new IP will not be updated automatically. To update the IP, you must remove the pool member and then add it again.

7. **Optional:** Set the pool member as backup member.

   Backup members will not receive traffic unless all other pool members are down.
8. **Optional:** Adjust the weight for the new pool member.

   A weight between 0 and 100 determines how much traffic the pool member will handle. The default weight is 1.

   - To deactivate the pool member, set the weight to 0 (no traffic will be routed to it).
   - To increase the traffic for the pool member, enter a weight between 1 and 100. Higher values will result in more traffic for the member.

     > **Note**
     >
     > Weights are uniformly mapped from the 0–255 range to the 0–100 range in the backend. This means that, for example, pool members with a weight of 1 and 2 will both be mapped to a weight of 1.

9. **Optional:** Enter tags to organize or classify the pool member.
10. **Optional:** To enable the pool member for traffic routing, select the **Admin State Up** checkbox.
11. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Pool Members

Delete pool members that you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** ▸ **Load Balancers**.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the pool member, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Pool Member JSON Configuration

View the JSON configuration of the pool member.

Procedure

1. Choose **Networking & Loadbalancing** ▸ **Load Balancers**.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the pool member, choose **JSON**.

   A dialog box opens and shows the pool member JSON configuration.

Creating TCP, PING, TLS-HELLO, and UDP Health Monitors

Create a health monitor that uses low-level network probes to check whether pool members respond to connection attempts.

Prerequisites

- You have at least one running server.

  For more information, see Servers.

- You have a load balancer.

  For more information, see Creating Load Balancers.

- You have a pool.

  For more information, see Creating Pools.

Context

Health monitors regularly check whether pool members are available and responsive. Based on the selected probe type, they test either basic connectivity or application-level behavior. If a member fails to respond as expected, it is marked as offline and removed from traffic distribution.

Use TCP, PING, TLS-HELLO, or UDP-CONNECT health monitors when you only need to check that a pool member accepts and responds to connections. These types do not inspect application-level content.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the pool to which you want to add a health monitor.
4. Choose New Health Monitor.

   A dialog box opens.
5. Enter a name.
6. Select the probe type:
   - PING
   - TCP
   - TLS-HELLO
   - UDP-CONNECT
7. Enter the maximum number retries before marking the pool members offline.
8. Enter the time interval in seconds between sending health check probes to pool members.
9. **Optional:** Enter tags to organize or classify the health monitor.
10. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Make sure that your security groups or firewall rules allow traffic from the health monitors' IP addresses to the pool members.

Related Information

[Audit](#)
[Security Groups](#)

Creating HTTP and HTTPS Health Monitors

Create a health monitor that uses HTTP or HTTPS probes to check the availability of pool members based on response codes and path checks.

Prerequisites

- You have at least one running server.

  For more information, see Servers.
- You have a load balancer.

For more information, see Creating Load Balancers.

- You have a pool.

  For more information, see Creating Pools.

Context

Health monitors regularly check whether pool members are available and responsive. Based on the selected probe type, they test either basic connectivity or application-level behavior. If a member fails to respond as expected, it is marked as offline and removed from traffic distribution.

Use HTTP or HTTPS health monitors to check application-level health. You can define expected status codes, request paths, and methods to verify that pool members serve valid application responses.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. Choose **New Health Monitor**.

   A dialog box opens.
5. Enter a name.
6. Select the probe type:
   - **HTTPS**
   - **HTTP**
7. Enter the maximum number of retries before marking the pool members offline.
8. Enter the time interval in seconds between sending health check probes to pool members.
9. Select the HTTP method:

Method	Description
CONNECT	Initiates a two-way communication tunnel (typically for TLS)
DELETE	Requests deletion of the specified resource
GET	Retrieves data without modifying the server
HEAD	Retrieves headers only, no body
OPTIONS	Returns the communication options for the target
PATCH	Applies partial modifications to a resource
POST	Sends data, typically to create resources
PUT	Replaces the resource at the target URL

10. Enter the expected HTTP status codes.

    You can use the following formats:
    - A single value, such as 200

- A comma-separated list, such as 200, 202
- A range, such as 200–204

11. Enter the URL path the health monitor requests.

    It must begin with a forward slash (/).

12. **Optional:** Enter tags to organize or classify the health monitor.
13. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Make sure that your security groups or firewall rules allow traffic from the health monitors' IP addresses to the pool members.

Related Information

Audit
Security Groups

Editing Health Monitors

Update the name, max retries, or interval of an existing health monitor.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **Edit**.

   A dialog box opens.

5. Enter a name.
6. Enter the maximum number of failed probes sent to the pool members before marking such pool members offline.
7. Enter the time interval in seconds between sending health check probes to pool members.
8. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Health Monitors

Permanently delete a health monitor you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Health Monitor JSON Configuration

View the JSON configuration of the health monitor.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **JSON**.

   A dialog box opens and shows the health monitor JSON configuration.

Domain Name System

Manage domain name system (DNS) zones and records.

Domain Name System Record Types

Get an overview of the domain name system (DNS) record types.

Sharing Domain Name System Zones with other Projects

Share domain name system (DNS) zones with other projects, but keep the records of the current project visible only to that project.

Deleting Domain Name System Zones

Delete domain name system (DNS) zones that you no longer need.

Creating Domain Name System Records

Create domain name system (DNS) records to define domain mappings, mail server settings, and other network configurations.

Editing Domain Name System Records

Update the content, description, and time-to-live of an existing domain name system (DNS) record.

Deleting Domain Name System Records

Delete domain name system (DNS) records that you no longer need.

Domain Name System Record Types

Get an overview of the domain name system (DNS) record types.

Record Type	**Description**
A	Internet Protocol version 4 (IPv4) address record Maps a domain to an IPv4 address.
AAAA	Internet Protocol version 6 (IPv6) address record Maps a domain to an IPv6 address.
CAA	Certification authority authorization record Specifies which certificate authorities can issue secure sockets layer (SSL) or transport layer security (TLS) certificates for the domain.
CERT	Certificate record Stores cryptographic certificates, including those used for SSL, TLS, and secure/multipurpose internet mail extensions (S/MIME).
CNAME	Canonical name record Creates an alias from one domain name to another.
HTTPS	HTTPS service binding record A specialized SVCB record for HTTP services. Allows clients to discover supported protocols, ports, and security-related parameters before connecting.
PTR	Pointer record Maps an internet protocol address to a domain name for reverse DNS lookups.
SPF	Sender policy framework record Specifies which mail servers are authorized to send e-mails on behalf of a domain, helping to prevent e-mail spoofing and phishing.
SRV	Service locator record Defines the hostname and port for specific services.
SVCB	Service binding record Provides connection details for a service endpoint, such as supported protocols, ports, alternative endpoints, and IP hints.
TXT	Text record Stores human-readable text data, commonly used for domain verification and security policies.

Sharing Domain Name System Zones with other Projects

Share domain name system (DNS) zones with other projects, but keep the records of the current project visible only to that project.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

- The DNS zone you want to share with another project must not already be shared with your project. After sharing, the target projects cannot share the zone with additional projects.

Procedure

1. Choose **Networking & Loadbalancing** › **DNS**.
2. From the gear dropdown menu for the zone, choose **Share this Zone**.

   A dialog box opens.
3. Enter a target project ID.
4. Chose **Share**.

Deleting Domain Name System Zones

Delete domain name system (DNS) zones that you no longer need.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** › **DNS**.
2. From the gear dropdown menu for the zone, choose **Delete**.

Creating Domain Name System Records

Create domain name system (DNS) records to define domain mappings, mail server settings, and other network configurations.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

- You know the supported record types.

  For more information, see Domain Name System Record Types.

Procedure

1. Choose Networking & Loadbalancing > DNS.
2. Choose the zone.
3. Choose **Create New**.

   A dialog box opens.
4. Select the record type.
5. **Optional:** Enter a name other than the zone name itself.
6. Based on the selected record type, enter the corresponding content.

   ### Example

   For A records, enter the IPv4 address.

   For CNAME records, enter the canonical name.
7. **Optional:** Enter a description.
8. **Optional:** Adjust the time-to-live (TTL) for the record, in seconds.
9. Choose **Create**.

Related Information

Floating IPs

Editing Domain Name System Records

Update the content, description, and time-to-live of an existing domain name system (DNS) record.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > DNS.
2. Choose the zone.
3. From the gear dropdown menu for the record, choose **Edit**.

   A dialog box opens.
4. Based on the record type, enter the corresponding content.

   ### Example

   For A records, enter the IPv4 address.

   For CNAME records, enter the canonical name.
5. **Optional:** Enter a description.
6. **Optional:** Adjust the time-to-live (TTL) for the record, in seconds.

7. Choose **Update**.

Deleting Domain Name System Records

Delete domain name system (DNS) records that you no longer need.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. Choose the zone.
3. From the gear dropdown menu for the record, choose **Delete**.

Storage

Manage storage capacity to save and manage your data.

Object Storage
Manage scalable storage for unstructured data such as backups or archives. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

File System Storage
Manage network-based storage for file sharing and file access. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

Container Image Registry
Manage container images using a centralized container image registry service to store and distribute them.

Object Storage

Manage scalable storage for unstructured data such as backups or archives. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

Object Storage Configuration Maximum
Get an overview of the maximum sizes for object storage.

Object Storage Throughputs
Get an overview of the benchmark results, which average multiple tests conducted under normal load conditions.

Creating Object Storage Containers
Create a object storage container to store objects within a project. Object storage containers enable you to define access permissions and set quotas for managing resources efficiently.

Uploading Objects to Object Storage Containers
Upload objects to your object storage containers to store data securely.

Downloading Objects from Object Storage Containers

Download objects from your object storage containers to retrieve stored data when needed.

Sharing Object Storage Containers Using Access Control

Control access to object storage containers by configuring permissions for secure sharing of stored data.

Deleting All Objects from an Object Storage Container

Delete all objects stored in an object storage container without deleting the container itself.

Deleting Object Storage Containers

Delete object storage containers that you no longer need.

Editing Object Storage Container Properties

Configure settings for an object storage container, such as quotas, public access, and metadata.

Editing Object Properties

Modify the properties of objects in your object storage, such as scheduling automatic deletion and defining metadata.

Object Storage Configuration Maximum

Get an overview of the maximum sizes for object storage.

Resource	Maximum Size
Number of Objects (per object storage container)	1,000,000 objects per object storage container
Object Size	5 GB per object Possible to upload files > 5 GB in chunks with segmentation

Object Storage Throughputs

Get an overview of the benchmark results, which average multiple tests conducted under normal load conditions.

Zone 2

File Size Range	Operation	Speed
10 MB – 100 MB	Upload	30–60 MB/s
10 MB – 100 MB	Download	50–74 MB/s

Zone 4

File Size Range	Operation	Speed
10 MB – 100 MB	Upload	20–28 MB/s
10 MB – 100 MB	Download	30–40 MB/s

Creating Object Storage Containers

Create a object storage container to store objects within a project. Object storage containers enable you to define access permissions and set quotas for managing resources efficiently.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for object storage.

  For more information, see Object Storage Configuration Maximum.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. Choose **Create container**.

   A dialog box opens.
3. Enter a name.
4. Choose **Save**.

Next Steps

- Uploading Objects to Object Storage Containers
- Editing Object Storage Container Properties

Uploading Objects to Object Storage Containers

Upload objects to your object storage containers to store data securely.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

- You have an object storage container.

  For more information, see Creating Object Storage Containers.

- You know and comply with the configuration limits for object storage.

  For more information, see Object Storage Configuration Maximum.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. Choose the object storage container.
3. Choose **Upload file**.

   A dialog box opens.
4. Select a file.
5. Enter a name.
6. Choose **Upload**.

Downloading Objects from Object Storage Containers

Download objects from your object storage containers to retrieve stored data when needed.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

- You have an object storage container with objects in it.

  For more information, see Creating Object Storage Containers.

- You know and comply with the configuration limits for object storage.

  For more information, see Object Storage Configuration Maximum.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. Choose the object storage container.
3. From the gear dropdown menu for the object, choose **Download**.

Sharing Object Storage Containers Using Access Control

Control access to object storage containers by configuring permissions for secure sharing of stored data.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. Choose the object storage container.
3. From the gear dropdown menu for the file, choose **Access Control**.

   A dialog box opens.

4. Specify the access control settings for your object storage container.
5. **Optional:** Choose **Check ACLs**.
6. Choose **Save**.

Deleting All Objects from an Object Storage Container

Delete all objects stored in an object storage container without deleting the container itself.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. From the gear dropdown menu for the object storage container, choose **Empty**.
3. Enter the container name to confirm the action.
4. Choose **Empty**.

Next Steps

Deleting Object Storage Containers

Deleting Object Storage Containers

Delete object storage containers that you no longer need.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

- The object storage container is empty.

  For more information, see Deleting All Objects from an Object Storage Container.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. From the gear dropdown menu for the object storage container, choose **Delete**.
3. Enter the container name.
4. Choose **Delete**.

Editing Object Storage Container Properties

Configure settings for an object storage container, such as quotas, public access, and metadata.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

- To edit or view static website serving properties, you've enabled public read access.

For more information, see Sharing Object Storage Containers Using Access Control.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. From the gear dropdown menu for the object storage container, choose **Properties**.
3. Edit the object storage container properties:

Option	Description
Object count quota	To limit the number of objects, enter a value. Leave this field empty to disable the quota.
Total size quota	To limit the total size of objects, enter a value. Leave this field empty to disable the quota.
URL for public access	Retrieve the URL for public access.
Serve objects as index	To serve objects as an index, select this checkbox and adjust the file name if needed.
Enable file listing	To list your object storage container's contents as a web page, select this checkbox.
Store old object versions	To store old versions of objects, select this checkbox.
Metadata	Set key-value pairs for metadata. Reserved keys include `web-index`, `quota-count`, and `quota-bytes`.

4. Choose **Update Container**.

Editing Object Properties

Modify the properties of objects in your object storage, such as scheduling automatic deletion and defining metadata.

Prerequisites

- You have the `objectstore_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared Object Storage**.
2. Choose the object storage container.
3. From the gear dropdown menu for the object, choose **Properties**.
4. Edit the object properties:

Option	Description
Expires at (UTC)	To schedule automatic deletion of the object, enter a timestamp.

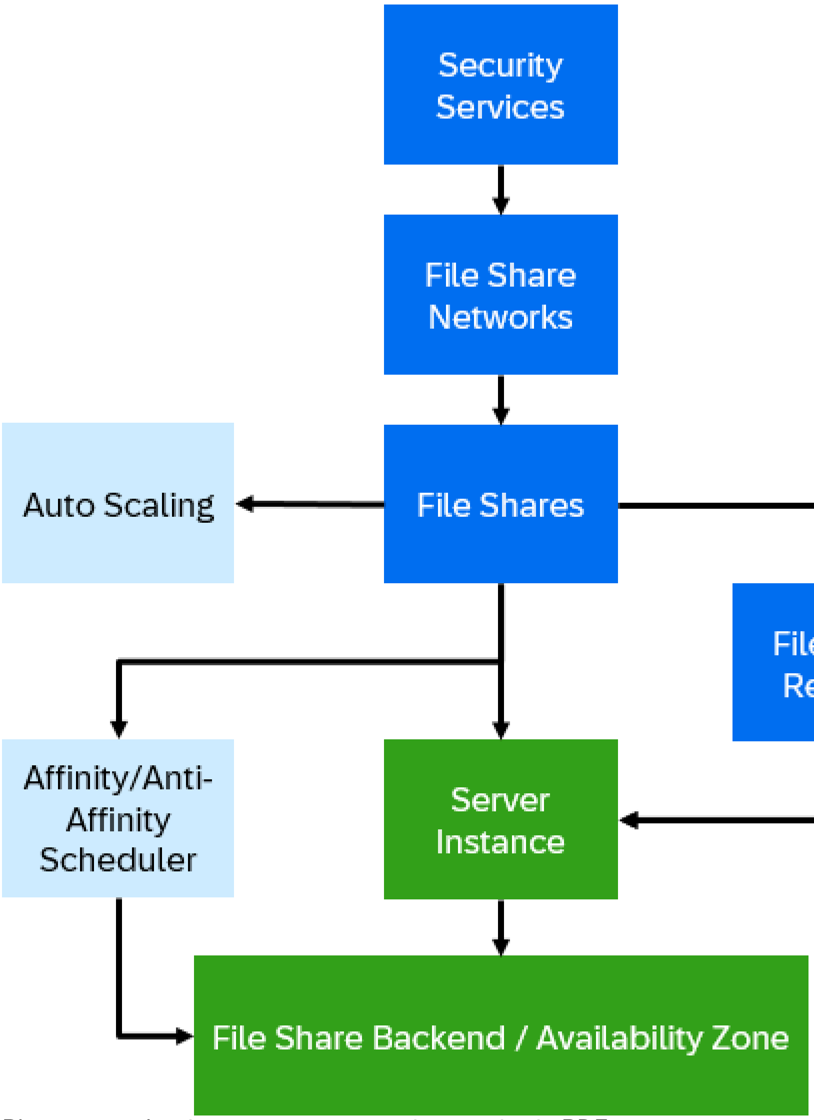
Option	Description
Metadata	Set key-value pairs for metadata.

5. Choose **Update object**.

File System Storage

Manage network-based storage for file sharing and file access. The service encrypts stored data automatically to reduce the risk of exposure if an unauthorized party gains physical access.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

[File System Storage Configuration Maximum](#)

Get an overview of the maximum sizes for file system storage.

[File System Storage Statuses](#)

Get an overview of the possible statuses for file shares, file share replicas, and security services.

[Security Services](#)

Security services connect your file shares to centralized authentication systems like LDAP or Active Directory. You can create, manage, and assign them to file share networks for secure access control.

[File Share Networks]

File share networks define the private network paths that connect file shares to your server instances. You can manage networks and control which security services they use.

[File Shares]

File shares provide shared file storage that multiple users or server instances can access over the network. You can create, manage, resize, and share file shares, as well as update their status and view error logs.

[Auto Scaling]

Manage and configure auto scaling for file shares, including enabling, monitoring activities, and disabling automatic resizing based on usage thresholds and conditions.

[File Share Replicas]

Replicas are copies of file shares created on different backends or in other availability zones. Use replicas to increase availability and improve data resilience.

[File Share Snapshots]

Snapshots capture the state of a file share at a specific point in time. You can use them to restore file shares, create new shares, or manage snapshot lifecycles.

File System Storage Configuration Maximum

Get an overview of the maximum sizes for file system storage.

Resource	Maximum Size
NFS/CIFS File	32 TB volume size

File System Storage Statuses

Get an overview of the possible statuses for file shares, file share replicas, and security services.

File Share Statuses

File Share Status	Description
creating	The file share is currently being created.
available	The file share is ready to use.
error	An error occurred. The file share cannot be used in its current state.
migrating	The file share is being moved to a different location or backend.
migrating_to	The file share is preparing to complete a migration process.
deleting	The file share is being deleted.
error_deleting	An error occurred during deletion.

File Share Replica Statuses

File Share Replica Status	Description
active	The replica is active and serves as the primary replica.
non-active	The replica is in standby and is not currently serving requests.
creating	The replica is being created.
deleting	The replica is being deleted.
error	An error occurred. The replica cannot be used in its current state.

File Share Snapshot Statuses

File Share Snapshot Status	Description
available	The snapshot is ready to use.
error	An error occurred. The snapshot cannot be used in its current state.
creating	The snapshot is being created.
deleting	The snapshot is being deleted.
error_deleting	An error occurred during deletion.
restoring	The snapshot is being used to restore a file share.

Security Services

Security services connect your file shares to centralized authentication systems like LDAP or Active Directory. You can create, manage, and assign them to file share networks for secure access control.

[Creating Active Directory Security Services](#)
Create an Active Directory (AD) security service to control access to file shares using your organization's AD domain. You can later add this security service to a file share network for centralized authentication.

[Creating LDAP Security Services](#)
Create an LDAP security service to control access to file shares using your organization's LDAP directory. You can later add this security service to a file share network for centralized authentication.

[Editing Security Services](#)
Update an existing security service.

[Deleting Security Services](#)
Permanently delete a security service you no longer need.

[Viewing Security Services Error Log](#)
View the error log of a security service.

Creating Active Directory Security Services

Create an Active Directory (AD) security service to control access to file shares using your organization's AD domain. You can later add this security service to a file share network for centralized authentication.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You use your own Active Directory and have configured it accordingly.

- The user you specify when creating the security service must have permission to create computer accounts (also called machine accounts or computer objects).

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Security Services** tab page.
3. Choose **Create New**.

   A dialog box opens.
4. Select the type: **Active Directory**
5. In the **OU (Organizational Unit)** field, enter the organizational unit that contains the AD users or groups you want to grant access to file shares.

   ### Example

   `OU=Users,DC=example,DC=com`

6. Enter a name to identify the security service when associating it with a file share network.
7. **Optional:** Enter a description.
8. **Optional:** Enter the IPv4 address of your AD DNS server.

   You can provide multiple IPv4 addresses, separated by commas.
9. **Optional:** In the **User** field, enter the username for the AD user that will bind to the directory.

   ### Example

   `username@domain.com`

10. **Optional:** Enter your AD domain name.
11. **Optional:** In the **Server** field, enter the IPv4 address of your preferred domain controller (DC).

    You can provide multiple IPv4 addresses, separated by commas.
12. Choose **Save**.

Results

- The Active Directory security service is created.

- The activity is recorded as an event in the audit log.

Next Steps

Adding Security Services to File Share Networks

Related Information

Creating LDAP Security Services

Create an LDAP security service to control access to file shares using your organization's LDAP directory. You can later add this security service to a file share network for centralized authentication.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Security Services** tab page.
3. Choose **Create New**.

   A dialog box opens.
4. Select the type: **LDAP**
5. In the **OU (Organizational Unit)** field, enter the base distinguished name (DN) of your LDAP directory.

   Do not include the actual organizational unit in this field.

   Example

   `dc=example,dc=com`
6. Enter a name to identify the security service when associating it with a file share network.
7. **Optional:** Enter a description.
8. **Optional:** Enter the IPv4 address of your LDAP DNS server.

   You can provide multiple IPv4 addresses, separated by commas.
9. **Optional:** In the **User** field, enter the username for the LDAP user that will bind to the directory.

   Example

   `username@domain.com`
10. **Optional:** Enter the domain name of your LDAP directory.
11. **Optional:** In the **Server** field, enter the IPv4 address of your preferred domain controller (DC).

    You can provide multiple IPv4 addresses, separated by commas.
12. Choose **Save**.

Results

- The LDAP security service is created.

- The activity is recorded as an event in the audit log.

Next Steps

Adding Security Services to File Share Networks

Related Information

Audit

Editing Security Services

Update an existing security service.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Security Services** tab page.
3. From the gear dropdown menu for the security service, choose **Edit**.

   A dialog box opens.
4. Edit the security service.

   If the security service is attached to a share network, you can only edit the **Description**, **User**, and **Password** fields.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Security Services

Permanently delete a security service you no longer need.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have removed the security service from all file share networks.

  For more information, see Removing Security Services from File Share Networks.

Procedure

1. Choose ▸ **Storage** ▸ **Shared File System Storage** ▸.
2. Choose the **Security Services** tab page.
3. From the gear dropdown menu for the security service, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing Security Services Error Log

View the error log of a security service.

Prerequisites

- You have the `sharedfilesystem_viewer` role.

  For more information, see [Authorizations].

Procedure

1. Choose ▸ **Storage** ▸ **Shared File System Storage** ▸.
2. Choose the **Security Services** tab page.
3. From the gear dropdown menu for the security service, choose **Error Log**.

   A dialog box opens and shows the error log entries.

Related Information

[Troubleshooting]

File Share Networks

File share networks define the private network paths that connect file shares to your server instances. You can manage networks and control which security services they use.

[Creating File Share Networks]

File share networks use your private networks to provide secure and isolated access to file shares. Create a file share network to connect your file shares to authorized server instances.

[Editing File Share Networks]

Update the name and description of an existing file share network.

[Deleting File Share Networks]

Permanently delete a file share network you no longer need.

[Viewing File Share Networks Error Log]

View the error log of a file share network.

[Adding Security Services to File Share Networks]

Add existing security services to a file share network to enable centralized authentication for your file shares.

[Removing Security Services from File Share Networks]

Delete security services from the file share network.

Creating File Share Networks

File share networks use your private networks to provide secure and isolated access to file shares. Create a file share network to connect your file shares to authorized server instances.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a private network.

  For more information, see Private Networks.

Procedure

1. Choose Storage > Shared File System Storage.
2. Choose the **Share Networks** tab page.
3. Choose **Create New**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Select the private network.
7. Select the subnet.
8. Choose **Save**.

Results

- Your file share network is created and listed on the **Share Networks** tab.
- The activity is recorded as an event in the audit log.

Next Steps

- If you want to use a security service (such as LDAP or Active Directory) with your file share network for enhanced security and access control, create and add a security service to your file share network.

  For more information, see:

  - Creating LDAP Security Services
  - Creating Active Directory Security Services

- Adding Security Services to File Share Networks

- Create a file share for the file share network.

  → Remember

  After you create a file share on this file share network, you cannot add a security service. Add any security services before creating a file share.

  For more information, see Creating File Shares.

Related Information

Audit

Editing File Share Networks

Update the name and description of an existing file share network.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Share Networks** tab page.
3. From the gear dropdown menu for the file share network, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting File Share Networks

Permanently delete a file share network you no longer need.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Share Networks** tab page.
3. From the gear dropdown menu for the file share network, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing File Share Networks Error Log

View the error log of a file share network.

Prerequisites

- You have the `sharedfilesystem_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Share Networks** tab page.
3. From the gear dropdown menu for the file share network, choose **Error Log**.

   A dialog box opens and shows the error log entries.

Related Information

Troubleshooting

Adding Security Services to File Share Networks

Add existing security services to a file share network to enable centralized authentication for your file shares.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a file share network that does not yet contain any file shares.

  For more information, see Creating File Share Networks.

- You have a security service.

  For more information, see:

  - Creating Active Directory Security Services
  - Creating LDAP Security Services

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Share Network** tab page.
3. From the gear dropdown menu for the file share, choose **Security Services**.

   A dialog box opens.

4. Choose **+**.
5. Select the security service.

   You can combine one Active Directory and one LDAP security service, or use just one of them.

6. Choose **Add**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating File Shares

Related Information

Audit

Removing Security Services from File Share Networks

Delete security services from the file share network.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have deleted all file shares from the file share network.

  → Remember

  The system updates the share network status once per hour. After deleting a file share, wait until the status is updated before removing the security service. If you try to remove the security service before the status is updated, you will see an error message: `Cannot remove security services. Share network is in use.`

For more information, see Deleting File Shares.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Share Network** tab page.
3. From the gear dropdown menu for the file share, choose **Security Services**.

   A dialog box opens.
4. Choose ×.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

File Shares

File shares provide shared file storage that multiple users or server instances can access over the network. You can create, manage, resize, and share file shares, as well as update their status and view error logs.

Affinity and Anti-Affinity Placement Strategies

Get an overview of affinity and anti-affinity strategies that are used by the file system storage scheduler to manage the placement of file shares on different backend clusters.

Configuring ID Mapping for NFSv4 File Shares

Configure ID mapping for NFSv4 file shares to ensure correct user ID mapping and prevent file access issues.

Creating File Shares

File shares are shared file systems that can be accessed by multiple users or instances. Create file shares to allow shared access and manage permissions for collaborative or central storage requests.

Editing File Shares

Update the name and description of an existing file share.

Deleting File Shares

Permanently delete a file share you no longer need.

Viewing File Shares Error Log

View the error log of a file share.

Sharing File Shares Using Access Control

Use access control to share file shares.

Changing File Share Sizes

Extend or shrink the size of a file share.

Changing File Share Status Manually

Manually update the status of a file share.

Viewing File Share Affinity and Anti-Affinity Metadata

View affinity and anti-affinity rules applied to file shares. These rules are stored as metadata when the share is created using the CLI or API.

Affinity and Anti-Affinity Placement Strategies

Get an overview of affinity and anti-affinity strategies that are used by the file system storage scheduler to manage the placement of file shares on different backend clusters.

Strategy	Scheduler Hint Field	Description	Recommended Max Set Size
Affinity	`share.scheduler_hints.same_host`	Places the new file share on the same backend cluster as specified file shares.	5 shares
Anti-Affinity	`share.scheduler_hints.different_host`	Places the new file share on a different backend cluster than specified file shares.	3 shares

You can create file shares with affinity or anti-affinity rules using the CLI or API.

Note

Affinity and anti-affinity rules are hard requirements. If the scheduler cannot satisfy the configured placement constraints, the file share will not be created and will enter an **error** state. This can happen for several reasons, such as:

- The referenced UUID does not exist.
- The placement cannot be fulfilled due to backend limitations.

Related Information

[Viewing File Share Affinity and Anti-Affinity Metadata]
[Viewing File Shares Error Log]
[Command-Line Guide for SAP Cloud Infrastructure]

Configuring ID Mapping for NFSv4 File Shares

Configure ID mapping for NFSv4 file shares to ensure correct user ID mapping and prevent file access issues.

Prerequisites

- You have root or appropriate administrative privileges on the system.

Context

NFSv4 uses an authentication domain to map local user IDs to user IDs on the NFS filer. If the ID mapping is not configured correctly, all file access will be mapped to the nobody user, which can cause issues when accessing files.

Procedure

1. Open the `/etc/idmapd.conf` file in an editor.

```
sudo nano /etc/idmapd.conf
```

The configuration file contains the following section:

```
[General]
Verbosity = 0
Pipefs-Directory = /run/rpc_pipefs
# set your own domain here, if it differs from FQDN minus hostname
# Domain = localdomain

[Mapping]
Nobody-User = nobody
Nobody-Group = nogroup
```

2. Uncomment and modify the **Domain** setting to use `defaultv4iddomain.com`:

```
[General]
Verbosity = 0
Pipefs-Directory = /run/rpc_pipefs
# set your own domain here, if it differs from FQDN minus hostname
Domain = defaultv4iddomain.com

[Mapping]
Nobody-User = nobody
Nobody-Group = nogroup
```

3. Save the file and restart the `rpcbind` service:

```
sudo systemctl restart rpcbind.service
```

4. Verify the user mapping by listing the files in the NFS share:

```
ls /fiotest/ -lah
```

Results

Once the configuration is updated and the service is restarted, the file ownership and permissions will display correctly, and users will no longer be mapped to the nobody user. The output of ls now shows the correct user IDs and groups.

Sample Code

```
root@nfsv4testserver:~# ls /fiotest/ -lah
total 8.0K
drwxr-xr-x  2 root root 4.0K Jan  9 09:30 .
drwxr-xr-x 23 root root 4.0K Aug 30 14:40 ..
-rw-r--r--  1 root root    0 Jan  9 09:30 testfile
```

→ Tip

If mapping is still incorrect after these changes, clear the ID mapping cache:

```
sudo nfsidmap -c
```

Next Steps

[Creating File Shares](#)

Creating File Shares

File shares are shared file systems that can be accessed by multiple users or instances. Create file shares to allow shared access and manage permissions for collaborative or central storage requests.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a file share network.

  For more information, see Creating File Share Networks.

- You know and comply with the configuration limits for file system storage.

  For more information, see File System Storage Configuration Maximum.

- **Optional:** To enable centralized authentication, you have added a security service to your file share network.

  For more information, see:

  - Creating Active Directory Security Services
  - Creating LDAP Security Services
  - Adding Security Services to File Share Networks

  > → **Remember**
  >
  > You can only add a security service before you create a file share on the file share network.

- **Optional:** To create an NFS file share using NFSv4, ensure that NFSv4 ID mapping is properly configured on your system. This configuration is necessary for correctly mapping user IDs between the client and the NFS server.

  For more information, see Configuring ID Mapping for NFSv4 File Shares.

- **Optional:** To create the file share based on a file share snapshot, ensure you have the UUID of the snapshot available.

  For more information, see File Share Snapshots.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. Choose **Create New**.

   A dialog box opens.
4. **Optional:** Enter a name.
5. **Optional:** Enter a description.
6. Select a protocol. You have the following options:

Protocol	Description
NFS	For UNIX/Linux environments using the Network File System protocol.

Protocol	Description
CIFS	For Windows environments using the SMB protocol with Active Directory integration.
MULTI	Supports both NFS and CIFS clients for hybrid environments; requires Active Directory and LDAP for authentication.

7. **Optional:** Select the type.
8. Enter the size in GB.
9. **Optional:** Enter the UUID of the file share's base snapshot.
10. **Optional:** Select the availability zone to determine the file share's location. If not specified, the service assigns one automatically.

    The service schedules file shares to file share backends based on availability zones.
11. Select the file share network.
12. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Mount the file share on one or more server instances.
- **Optional:** Sharing File Shares Using Access Control

Related Information

Audit

Editing File Shares

Update the name and description of an existing file share.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Edit**.

   A dialog box opens.
4. Enter a name.

5. Enter a description.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting File Shares

Permanently delete a file share you no longer need.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have deleted all snapshots of the file share.

  For more information, see Deleting File Share Snapshots.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Delete**.
4. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- If you plan to remove a security service from the file share network, keep in mind that the system updates the network status every hour. Trying to remove the security service before the update completes results in the following error message: `Cannot remove security services. Share network is in use.`

  For more information, see Removing Security Services from File Share Networks.

Related Information

Audit

Viewing File Shares Error Log

View the error log of a file share.

Prerequisites

- You have the `sharedfilesystem_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Error Log**.

   A dialog box opens and shows the error log entries.

Related Information

Troubleshooting

Sharing File Shares Using Access Control

Use access control to share file shares.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You know the fixed IP address (range) of the server to which you want to grant access to this share.

- The file share is in status **available**.

  For more information, see File Share Statuses.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Access Control**.

   A dialog box opens.

4. Choose **+**.
5. Select the access type **ip**.
6. Enter the fixed IP address (range).
7. Select the access level. You have the following options:

   - **read only**
   - **read-write**

8. Choose **Save**.

Changing File Share Sizes

Extend or shrink the size of a file share.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Context

→ Recommendation

Before extending or shrinking a file share, ensure backups exist and a rollback plan is in place. Verify data integrity before performing a shrink operation.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Extend / Shrink**.

   A dialog box opens.
4. Enter the size in GB.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit
Auto Scaling

Changing File Share Status Manually

Manually update the status of a file share.

Prerequisites

- You have the `admin` role.

  For more information, see Authorizations.

- You know the possible file share statuses.

For more information, see File Share Statuses.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Reset Status**.

   A dialog box opens.
4. Select the status.
5. Choose **Save**.

Results

- The file share status is updated to the selected value.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing File Share Affinity and Anti-Affinity Metadata

View affinity and anti-affinity rules applied to file shares. These rules are stored as metadata when the share is created using the CLI or API.

Prerequisites

- You have created file shares with affinity or anti-affinity rules using the CLI or API.

Context

Affinity and anti-affinity rules are stored as metadata and cannot be modified after file share creation.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. Choose the share.

   A dialog box opens.
4. Choose the **Metadata** tab page.

Results

You can now see which affinity or anti-affinity rules were applied to the share. The metadata values include:

- **__affinity_same_host**: UUIDs of shares that are colocated.

- **__affinity_different_host**: UUIDs of shares that are placed on different hosts.

These metadata entries reflect the scheduler hints that were used at creation time.

Next Steps

If a share failed to schedule due to affinity or anti-affinity settings, check the error log for more details.

Related Information

Viewing File Shares Error Log

Auto Scaling

Manage and configure auto scaling for file shares, including enabling, monitoring activities, and disabling automatic resizing based on usage thresholds and conditions.

Enabling or Editing Auto Scaling
Resize file shares automatically based on configured usage thresholds or other conditions.
Viewing Auto Scaling Activities and Errors
View auto scaling activities and any errors that occur during file share resizing.
Disabling Auto Scaling
Disable auto scaling for file shares to stop automatic resizing based on usage thresholds and conditions.

Enabling or Editing Auto Scaling

Resize file shares automatically based on configured usage thresholds or other conditions.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- If you need to exclude specific file shares from auto scaling, you have set the `snapmirror` metadata key to value `1` using the command-line.

  For more information, see Command-Line Guide for SAP Cloud Infrastructure.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Autoscaling** tab page.
3. Choose **Configure**.

   A dialog box opens.
4. Configure actions for low usage.
   a. Select an action:

- Do nothing

  Keep the current file share size.

- Auto-shrink share

  Reduce the file share size automatically.

b. Enter the observation period in minutes (default: 180).

c. Enter the usage threshold in percent (default: 50%).

When usage stays below this threshold for the specified duration, the selected action is triggered.

5. Configure actions for high usage.

a. Select an action:

- Do nothing

  Keep the current file share size.

- Auto-extend share

  Increase the file share size automatically.

b. Enter the observation period in minutes (default: 60).

c. Enter the usage threshold in percent (default: 80%).

When usage exceeds this threshold for the specified duration, the selected action is triggered.

6. Configure actions for critical usage.

a. Select an action:

- Do nothing

  Keep the current file share size.

- Auto-extend share immediately

  Increase the file share size immediately.

b. Enter the critical usage threshold in percent (default: 95%).

When usage exceeds this critical threshold, the selected action is triggered immediately.

7. Select the stepping strategy:

Stepping Strategy	Description
Percentage-step resizing	Resize the file share by a percentage of its current size during each operation. ◦ When the critical threshold is crossed, auto scaling can take multiple steps at once to quickly reduce usage. ◦ Recommended if usage changes frequently. This option is a good default for most shared file system shares.
Single-step resizing	Resize the file share in a single step to return usage to normal levels. ◦ When both critical and high thresholds are crossed, auto scaling calculates a new size to leave both thresholds.

Stepping Strategy	Description
	◦ Recommended if usage changes infrequently but in large steps.

8. Enter a minimum total size in GiB to prevent scaling below this limit.
9. Enter a maximum total size in GiB to prevent scaling above this limit.
10. Enter the amount of free space to maintain in GiB.

    When usage drops below this free space threshold, resizing occurs immediately to maintain free space.
11. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Viewing Auto Scaling Activities and Errors

Related Information

Metrics Catalog
Audit

Viewing Auto Scaling Activities and Errors

View auto scaling activities and any errors that occur during file share resizing.

Prerequisites

- You have the `sharedfilesystem_viewer` role.

  For more information, see Authorizations.

- You have enabled auto scaling.

  For more information, see Enabling or Editing Auto Scaling.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Autoscaling** tab page.
3. Review the following sections:

Section	Description
Configuration	Shows the current auto scaling settings for file shares.
Pending operations	Shows scheduled auto scaling activities.

Section	Description
Recently succeeded	Shows completed auto scaling activities.
Recently failed	Shows failed auto scaling activities.
Scraping errors	Shows errors that occurred while collecting usage data for auto scaling decisions.

Related Information

[Metrics Catalog](#)

Disabling Auto Scaling

Disable auto scaling for file shares to stop automatic resizing based on usage thresholds and conditions.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have enabled auto scaling.

  For more information, see Enabling or Editing Auto Scaling.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Autoscaling** tab page.
3. Choose **Disable autoscaling**.
4. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

File Share Replicas

Replicas are copies of file shares created on different backends or in other availability zones. Use replicas to increase availability and improve data resilience.

Creating File Share Replicas

Creating File Share Replicas

Create replicas of a file share, known as file share instances, on specific backends. You can create several replicas to guarantee the presence of the instance in various availability zones.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a file share.

  For more information, see Creating File Shares.

- The file share is in status **available**.

  For more information, see File Share Statuses.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Create Replica**.

   A dialog box opens.
4. Select the availability zone.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Note

To manage file share replicas after creation, use the CLI.

Related Information

Audit
Command-Line Guide for SAP Cloud Infrastructure

File Share Snapshots

Snapshots capture the state of a file share at a specific point in time. You can use them to restore file shares, create new shares, or manage snapshot lifecycles.

[Creating File Share Snapshots]
Create a copy of a file share at a specific point in time. You can use a snapshot to create new file shares with the snapshot data or to revert an existing share.
[Editing File Share Snapshots]
Update the name and description of an existing file share snapshot.
[Reverting File Shares Using Snapshots]
Restore a file share to a previous state using an existing snapshot.
[Deleting File Share Snapshots]
Permanently delete a file share snapshot you no longer need.
[Viewing File Share Snapshots Error Log]
View the error log of a file share snapshot.

Creating File Share Snapshots

Create a copy of a file share at a specific point in time. You can use a snapshot to create new file shares with the snapshot data or to revert an existing share.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a file share.

  For more information, see Creating File Shares.

- The file share is in status **available**.

  For more information, see File Share Statuses.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Create Snapshot**.

   A dialog box opens.
4. Enter a name.
5. Enter a description.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing File Share Snapshots

Update the name and description of an existing file share snapshot.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Snapshots** tab page.
3. From the gear dropdown menu for the file share snapshot, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. Enter a description.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Reverting File Shares Using Snapshots

Restore a file share to a previous state using an existing snapshot.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see Authorizations.

- You have a snapshot.

  For more information, see Creating File Share Snapshots.

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Shares** tab page.
3. From the gear dropdown menu for the file share, choose **Revert To Snapshot**.

   A dialog box opens.

4. Select the snapshot.
5. Choose **Revert**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Deleting File Share Snapshots

Permanently delete a file share snapshot you no longer need.

Prerequisites

- You have the `sharedfilesystem_admin` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Snapshots** tab page.
3. From the gear dropdown menu for the file share snapshot, choose **Delete**.
4. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing File Share Snapshots Error Log

View the error log of a file share snapshot.

Prerequisites

- You have the `sharedfilesystem_viewer` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Storage** > **Shared File System Storage**.
2. Choose the **Snapshots** tab page.
3. From the gear dropdown menu for the file share snapshot, choose **Error Log**.

   A dialog box opens and shows the error log entries.

Related Information

Troubleshooting